MIL-STD-1232 (ORD)
20 April 1961

MILITARY STANDARD

VISUAL INSPECTION STANDARDS FOR CUPS AND DISKS USED IN SMALL ARMS AMMUNITION MANUFACTURE

UNITED STATES
GOVERNMENT PRINTING OFFICE
WASHINGTON : 1962

FSC 1305

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

FOREWORD

This standard was prepared under the direction of the Office, Chief of Ordnance to be used to supplement specifications for cups and disks used in the manufacture of small arms ammunition. It replaces the following publication on the same subject: ORD–SIP–S301, 1 July 1952.

This publication is a military standard which defines and illustrates a practical majority of defects assembled as a result of extensive surveys covering cup, disk, and small arms ammunition manufacturing facilities.

It is realized that photographs themselves impose a limitation; being two-dimensional, they may not indicate effectively variations in depth which should be considered in classifying defects such as scratches and dents.

The format retains most of the features of the format of the publication it replaces.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CONTENTS

		Page
1.	SCOPE	1
1.1	Purpose	1
1.2	Scope	1
2.	REFERENCE DOCUMENTS	1
3.	DEFINITIONS	1
3.1	Specific	1
4.	GENERAL REQUIREMENTS	1
4.1	General	1
5.	DETAIL REQUIREMENTS	2
5.1	Classification	2
5.1.1	Defect Grouping	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 PURPOSE. The purpose of this standard is to:

(a) Create standardization of defect nomenclatures and establish the degree of nonconformity, which constitutes defectiveness.

(b) Provide visual inspection standards for use in connection with acceptance inspection of cups and disks.

(c) Serve as an instruction manual in the training of personnel engaged in the manufacture and inspection of cups and disks.

1.2 SCOPE. This standard provides procedures for determining and evaluating defects in cups and disks used in the manufacture of small arms ammunition, and the standard shall be used as a supplement to pertinent specifications.

2. REFERENCED DOCUMENTS

(None)

3. DEFINITIONS

3.1 SPECIFIC. For the purpose of this standard, the following definitions shall apply:

3.1.1 *Unacceptable.* A condition considered undesirable and categorized as a defect either because it could result in failure or materially reduce the usability of the item, or because it represents a significant departure from established standards of workmanship. Classification of such defects as major or minor shall be in accordance with the detail specification.

3.1.2 *Acceptable.* A condition which, though not desirable from an appearance standpoint, is permitted for reasons of economy.

4. GENERAL REQUIREMENTS

4.1 GENERAL. The descriptions and definitions of defects established in this standard are those commonly occurring. Any other defects not listed, present in the lot, shall be reported. If, in the opinion of the Government inspector, the other defects be serious, complete information, recommendations, and request for disposition shall be submitted through proper channels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

5. DETAIL REQUIREMENTS

5.1 CLASSIFICATION

5.1.1 *Defect Grouping.* In some instances, the standards for one type of cup are also applicable to another type. Where this occurs the applicability is so indicated. For convenience in referencing, the defect photographs are grouped as follows:

	Page
Caliber .30 Brass Case Cups (Four Draw)	3
Caliber .30 Brass Case Cups (Two Draw)	7
NATO 7.62-MM Brass Case Cups (Two Draw)	7
NATO 7.62-MM Brass Case Cups (Three Draw)	10
Caliber .30 Brass Case Cups (Three Draw)	10
Caliber .30 Steel Case Cups	13
Caliber .30 Carbine Brass Case Cups	16
Caliber .30 Gilding Metal Bullet Jacket Cups	18
Caliber .30 Gilding Metal Clad Steel Bullet Jacket Cups	20
Caliber .30 Carbine Gilding Metal Bullet Jacket Cups	22
Caliber .30 Carbine Steel Case Cups	24
Caliber .45 Steel Case Cups	24
Caliber .45 Gilding Metal Clad and Copper Plated Steel Bullet Jacket Cups	26
Caliber .50 Brass Case Cups (Three Draw)	28
Caliber .50 Brass Case Cups (Two Draw)	32
Caliber .50 Steel Case Cup Disks	36
Caliber .50 Steel Case Cups	41
Caliber .50 Gilding Metal Bullet Jacket Cups	45
Caliber .50 Gilding Metal Clad Steel Bullet Jacket Cups	48
20-MM Brass Case Cup Disks	52
20-MM Brass Case Cups	57

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .30 BRASS CASE CUPS (FOUR DRAW)



FIGURE 1. *Scaly metal—unacceptable.*

Any cup which appears, without magnification, to be scaly is to be considered a defective. Scaly metal includes slivers, blisters, wire edges, and breaks. Wire edges may occur on either the inner or outer wall of the cup. Cups captioned "Scaly Metal" figures 15 and 30 and identified as A, B, C, and A, B, C, respectively, are examples of unacceptable wire edges; those cups identified as D, E, and D, E, F, figures 16 and 31, respectively, are acceptable.



FIGURE 2. *Improperly formed or mutilated—unacceptable.*

Cups having "V'd" edges, spills, etc., as well as those mutilated by pickling equipment, conveyors, or rough handling are included in this group.



FIGURE 3.—*Improperly formed or mutilated—acceptable.*



FIGURE 4. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives.



FIGURE 5. *Scratches—acceptable.*

Light scratches on the interior or exterior base which fade out in the sidewall and which have clean edges are acceptable. Draw scratches on the sidewall if no worse than in the cup at the right end, are acceptable as are light interior sidewall scratches which do not carry across the base.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

Lamination has been a troublesome cup defect. In order to determine whether or not suspected laminations are to be counted as defects, cups are to be sectioned, polished,

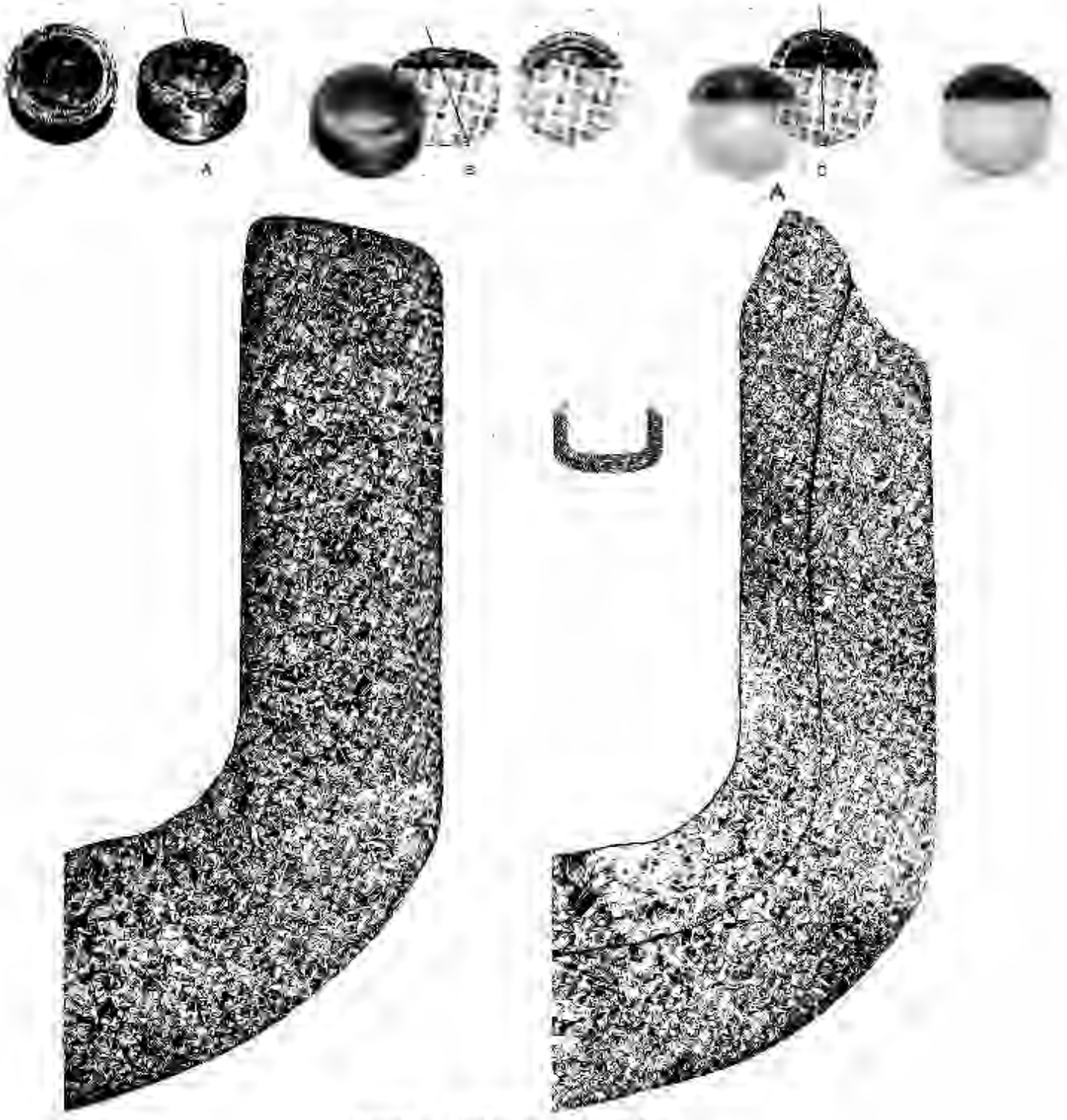


FIGURE 6. *Laminations—Unacceptable*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

etched and examined with a microscope. A cup is to be called laminated if a break extends more than .020″ below an imaginary line drawn through the lowest point of the mouth bevel, across the sidewall and perpendicular to the cup axis. Eight cups are shown which were suspected of being laminated. Microscopic examination after the cups were sectioned, polished and etched indicated that all were defective. Microphotographs (13 times actual size) of three cups marked A, B, and C, together with that of a good cup (extreme left below) are shown. In this booklet only laminated caliber .30 brass case cups are shown because of space limitations. What is written here applies equally to other cups. In addition, in steel case cups, seams and cracks should be examined after sectioning, polishing and etching. Steel case cups are defective if seams or cracks extend into the body of the metal.

A disk is to be called laminated if a break extends more than .015″ beyond (toward disk center) an imaginary line drawn perpendicular to the disk through a point on the periphery of its minimum diameter.

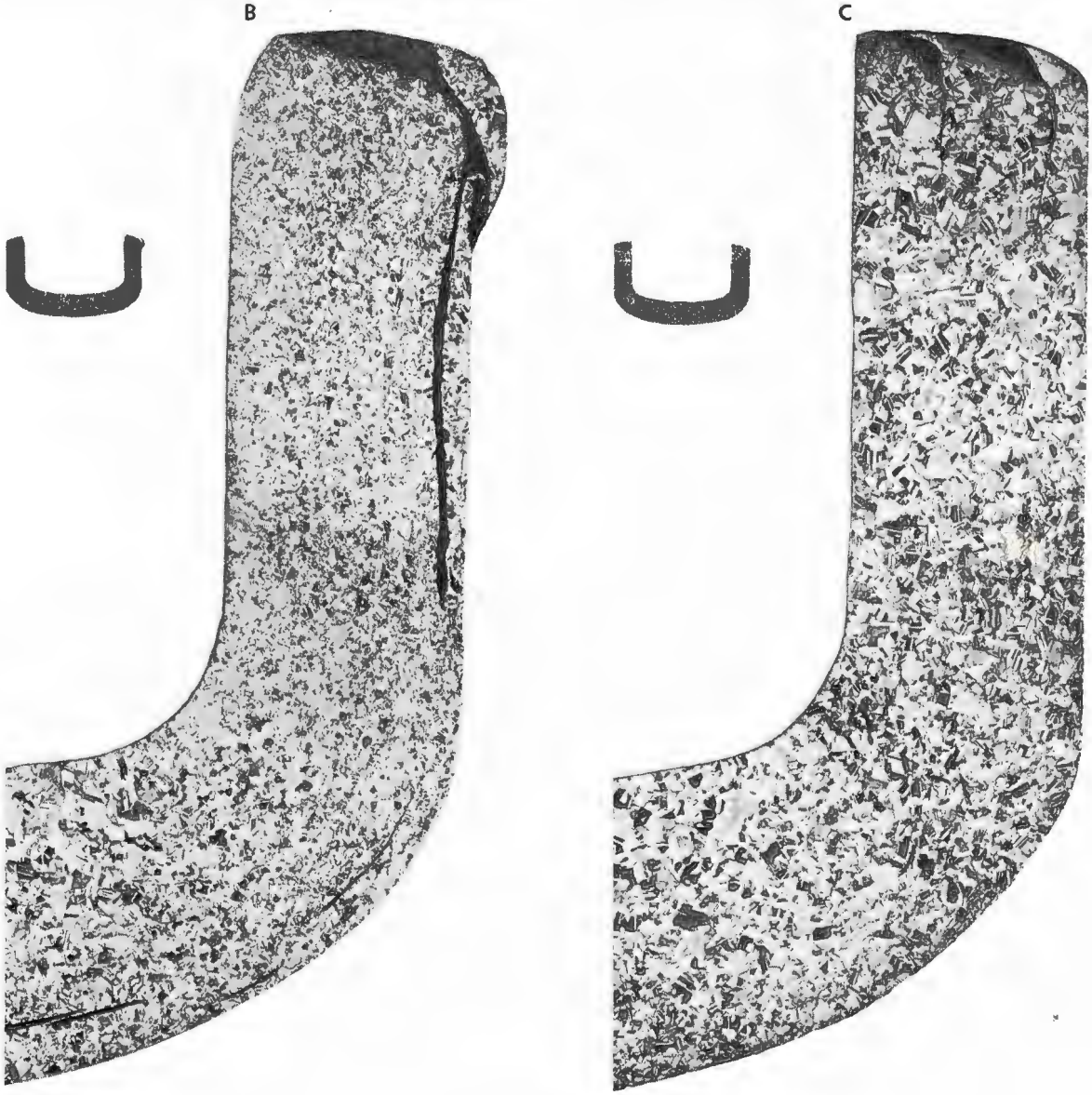


FIGURE 6.—Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 7. *Oil, grease, dirt—unacceptable.*



FIGURE 8. *Oil, grease, dirt—acceptable.*



FIGURE 9. *Oxidation—unacceptable.*
Discoloration like that illustrated, caused by failure in pickling and rinsing to remove coatings formed in annealing or when hot cups are immersed in pickling solution, unacceptable.



FIGURE 10. *Oxidation—acceptable.*
The bright yellow color produced by over-pickling is to be considered acceptable oxidation, even though it is undesirable.



FIGURE 11. *Red spots—unacceptable.*
Defectives are cups having copper deposited on them due to electrochemical action in pickling. Such action generally occurs when free iron is present. Black specks in the red spots indicate iron inclusions. Cups with such inclusions and those with large areas of red discoloration are defectives.



FIGURE 12. *Red spots—acceptable.*
Cups are not defectives where pickling action has removed inclusions, or where area affected is no greater than that shown.

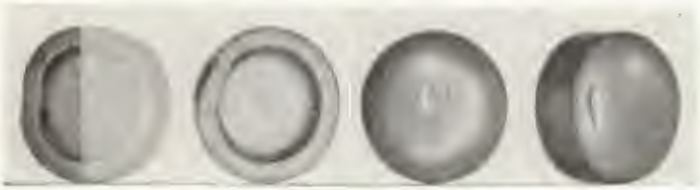


FIGURE 13. *Dents or bent edges—unacceptable.*



FIGURE 14. *Dents or bent edges—acceptable.*
If cups are not bent out-of-round or dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .30 BRASS CASE CUPS (TWO DRAW)
Standards Applicable Also to
NATO 7.62-MM BRASS CASE CUPS (TWO DRAW)



FIGURE 15. *Scaly metal—unacceptable.*



FIGURE 16. *Scaly metal—acceptable.*

Any cup which appears, without magnification, to be scaly is to be considered a defective. Scaly metal includes slivers, blisters, wire edges, and cracks. Wire edges may occur on either the inner or outer wall of the cup. Cups identified as A, B, C are examples of unacceptable wire edges; cups D, E are acceptable. Note slight sliver mark on cup identified as F.



FIGURE 17. *Improperly formed or mutilated—unacceptable.*

Cups having "V'd" edges, spills, etc., as well as those mutilated by pickling equipment, conveyors, or rough handling are included in this group.

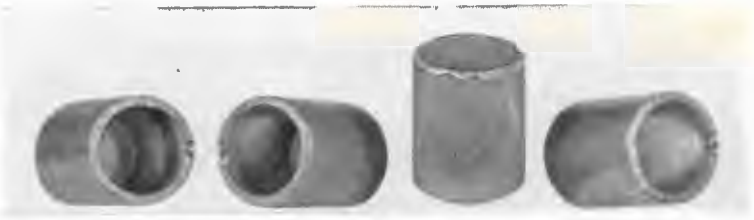


FIGURE 18. *Improperly formed or mutilated—acceptable.*

Laminations—unacceptable (not illustrated.) (See fig. 6.)



FIGURE 19. *Scrap—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 20. *Scratches—unacceptable.*
Cups having deep scratches such as those shown are defectives.



FIGURE 21. *Scratches—acceptable.*
Light scratches on the interior (see acceptable scratches on Cal. .30 Brass Case Cups, fig. 5) or exterior base which fade out in the sidewall and which have clean edges are acceptable. Draw scratches on the sidewall, if no worse than in #2 cup, are acceptable as are light interior sidewall scratches which do not carry across the base.

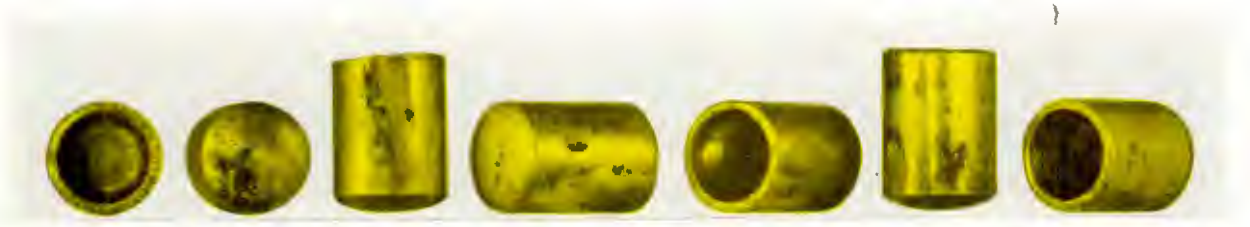


FIGURE 22. *Oil, grease, dirt—unacceptable.*



FIGURE 23. *Oil, grease, dirt—acceptable.*



FIGURE 24. *Oxidation—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 25. *Oxidation—acceptable.*

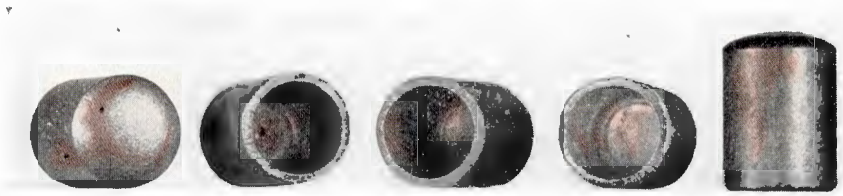


FIGURE 26. *Red spots—Unacceptable.*

Defectives are cups having copper deposited on them due to electrochemical action in pickling. Such action generally occurs when free iron is present. Black specks in the red spots indicate iron inclusions. Cups with such inclusions and those with large areas of rod discoloration are defectives.



FIGURE 27. *Red spots—acceptable.*

Cups are not defectives where pickling action has removed inclusions, or where area affected is no greater than that shown.



FIGURE 28. *Dents or bent edges—unacceptable.*



FIGURE 29. *Dents or bent edges—acceptable.*

If cups are not bent out-of-round or dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

NATO 7.62-MM BRASS CASE CUPS (THREE DRAW)
Standards Applicable Also to
CALIBER .30 BRASS CASE CUPS (THREE DRAW)

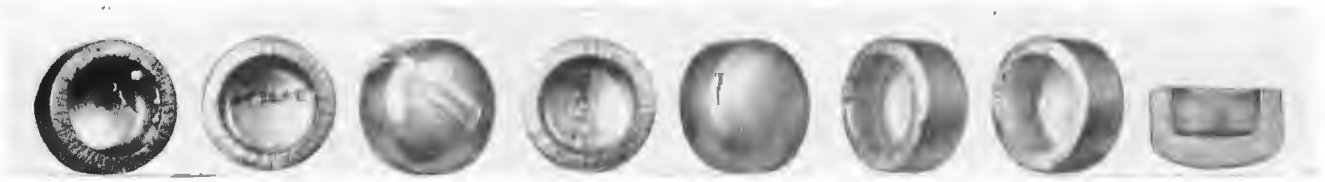


FIGURE 30. *Scaly metal—unacceptable.*



FIGURE 31. *Scaly metal—acceptable.*

Any cup which appears, without magnification, to be scaly is to be considered a defective. Scaly metal includes slivers, blisters, wire edges, and cracks. Wire edges may occur on either the inner or outer wall of the cup. Cups identified as A, B, C are examples of unacceptable wire edges; cups identified D, E, F are acceptable.



FIGURE 32. *Improperly formed or mutilated—unacceptable.*

Cups having "V'd" edges, spills, etc., as well as those mutilated by pickling equipment, conveyors, or rough handling are included in this group.



FIGURE 33. *Improperly formed or mutilated—acceptable.*

•

Laminations—unacceptable (Not illustrated.) (See Fig. 6.)

•



FIGURE 34. *Scrap—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 35. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives.



FIGURE 36. *Scratches—acceptable.*

Light scratches on the interior or exterior base which fade out in the sidewall and which have clean edges are acceptable. Draw scratches, if no worse than in cup #5 above, are acceptable as are light interior sidewall scratches which do not carry across the base.

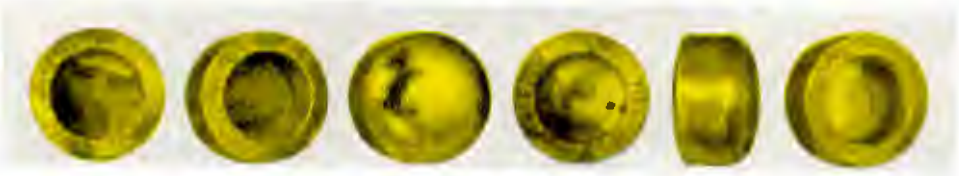


FIGURE 37. *Oil, grease, dirt—unacceptable.*



FIGURE 38. *Oil, grease, dirt—acceptable.*

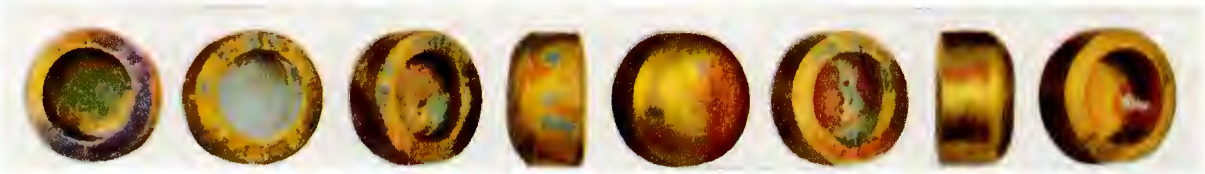


FIGURE 39. *Oxidation—unacceptable.*



FIGURE 40. *Oxidation—acceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

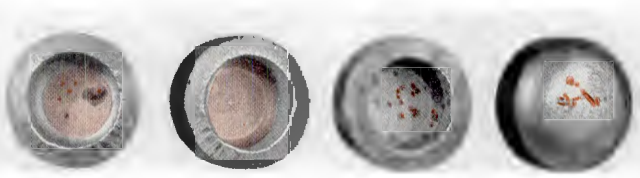


FIGURE 41. *Red spots—unacceptable.*

Defectives are cups having copper deposited on them due to electrochemical action in pickling. Such action generally occurs when free iron is present. Black specks in the red spots indicate iron inclusions. Cups with such inclusions and those with large areas of red discoloration are defectives.



FIGURE 42. *Red spots—acceptable.*

Cups are not defectives where pickling action has removed inclusions, or where area affected is relatively small.

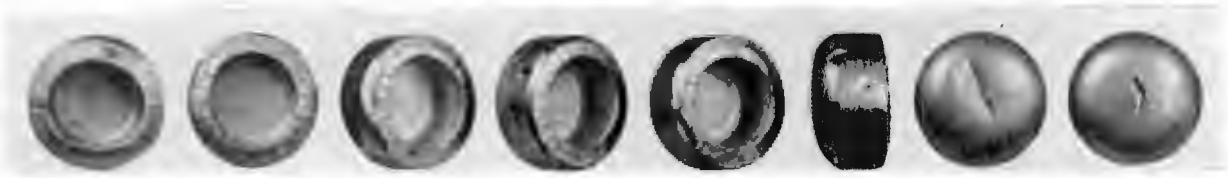


FIGURE 43. *Dents or bent edges—unacceptable.*

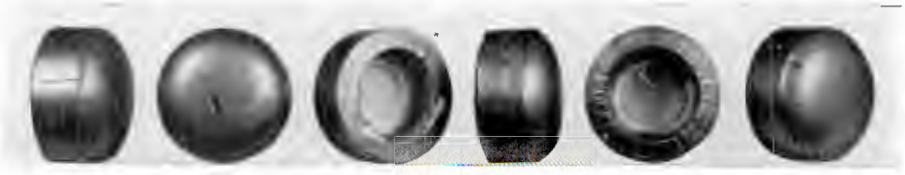


FIGURE 44. *Dents or bent edges—acceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .30 STEEL CASE CUPS



FIGURE 45. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives.



FIGURE 46. *Scratches—acceptable.*



FIGURE 47. *Improperly formed or mutilated—unacceptable.*

This group includes cups having "V'd" edges, spills, etc., as well as those mutilated by rough handling or equipment.



FIGURE 48. *Improperly formed or mutilated.*

Small spills like those shown are acceptable. "V'd" edges like that shown shall be unacceptable.

•

Laminations—unacceptable. (Not illustrated.) (See Fig. 6.)

•



FIGURE 49. *Scrap—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 50. *Scaly metal—unacceptable.*

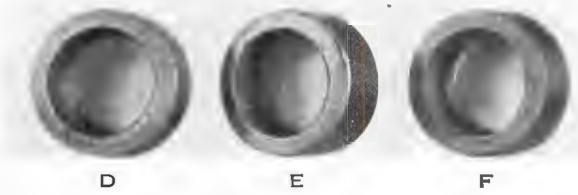


FIGURE 51. *Scaly metal—acceptable.*

A cup which appears, without magnification, to have been scaly before it was copper coated is to be counted as defective. Scaly metal includes wire edges, slivers, blisters, and cracks, as well as coating which has been drawn up into a wire edge. Wire edges may occur on either the inner or outer wall of the cup. Cups identified as A, B, C are examples of unacceptable wire edges; cups identified D, E, F are acceptable.

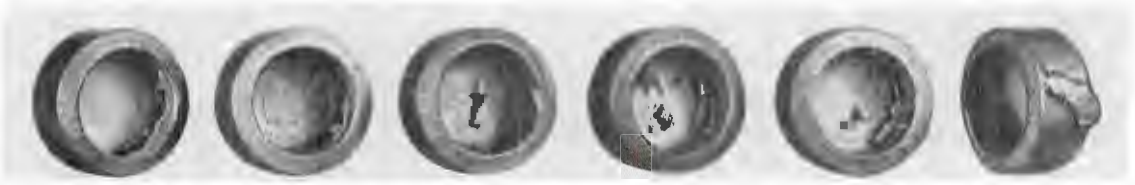


FIGURE 52. *Nonadherent copper coating—unacceptable*

A cup from which the copper coating is peeling or on which the coating has blistered is to be counted as a defective.



FIGURE 53. *Nonadherent copper coating—acceptable.*

Note lesser degree of defect.



FIGURE 54. *Oil, grease, dirt—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 55. *Oil, grease, dirt—acceptable.*

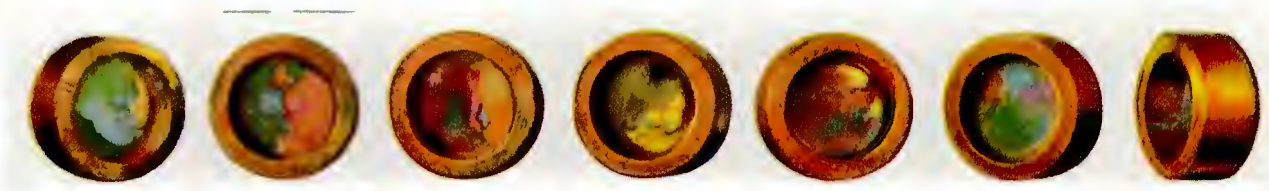


FIGURE 56. *Oxidation—unacceptable.*

Oxidation includes discoloration because of failure in pickling and rinsing to remove the oxides formed in annealing or when hot cups are dipped in pickling solution.



FIGURE 57. *Oxidation—acceptable.*



FIGURE 58. *Dents or bent edges—unacceptable.*



FIGURE 59. *Dents or bent edges—acceptable.*

If cups are not dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .30 CARBINE BRASS CASE CUPS



FIGURE 60. *Scaly metal—unacceptable.*

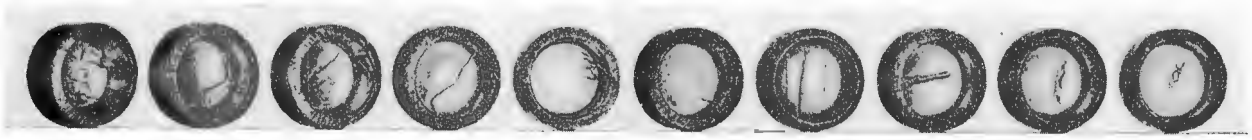


FIGURE 61. *Scaly metal—unacceptable.*

Any cup which appears, without magnification, to be scaly is to be considered a defective. Scaly metal includes slivers, blisters, wire edges, and cracks. Wire edges may occur on either the inner or outer wall of the cup. Cups captioned "Scaly Metal" figures 15 and 30 and identified as A, B, C, and A, B, C, respectively, are examples of unacceptable wire edges; those cups identified as D, E, and D, E, F, figures 16 and 31, respectively, are acceptable.



FIGURE 62. *Improperly formed or mutilated—unacceptable.*

Cups having "V'd" edges, spills, etc., as well as those mutilated by pickling equipment, conveyors or rough handling are included in this group.



FIGURE 63. *Improperly formed or mutilated—acceptable.*

Spills no deeper than the "V'd" edges shown are not to be counted as defectives.

•

Laminations—unacceptable. (Not illustrated.) (See Fig. 6.)

•



FIGURE 64. *Scratches—unacceptable.*

Cups having deep scratches sueh as those shown are defectives.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

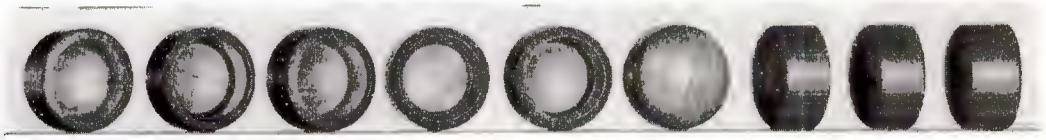


FIGURE 65. *Scratches—acceptable.*

Light scratches on the interior or exterior base which fade out in the sidewall and light interior sidewall scratches which do not carry across the base, are acceptable if they have clean edges. Exterior draw scratches, if no worse than those on the last three cups at the right, are acceptable.



FIGURE 66. *Oil, grease, dirt—unacceptable.*

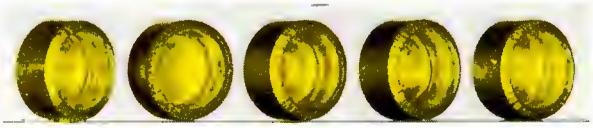


FIGURE 67. *Oil, grease, dirt—acceptable.*

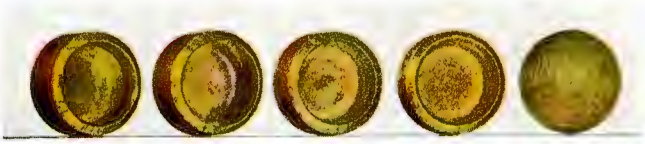


FIGURE 68. *Oxidation—unacceptable.*

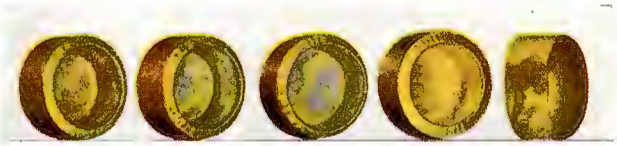


FIGURE 69. *Oxidation—acceptable.*



FIGURE 70. *Red spots—unacceptable.*

Defectives are cups having copper deposited on them due to electrochemical action in pickling. Such action generally occurs when free iron is present. Black specks in the red spots indicate iron inclusions. Cups with such inclusions and those with large areas of red discoloration are defectives.



FIGURE 71. *Red spots—acceptable.*

Cups are not defectives where pickling action has removed inclusions, or where area affected is relatively small.



FIGURE 72. *Dents or bent edges—unacceptable.*



FIGURE 73. *Dents or bent edges—acceptable.*

If cups are not bent out-of-round or dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .30 GILDING METAL BULLET JACKET CUPS



FIGURE 74.—*Scaly metal—unacceptacle.*

Any cup which appears, without magnification, to be scaly is to be counted as a defective. Scaly metal includes slivers, blisters, cracks, and wire edges. Wire edges on the inner or outer wall of the cup (not shown above) shall be unacceptable.



FIGURE 75.—*Improperly formed or mutilated—unacceptable.*

Cups having "V'd" edges, spills, etc., as well as those mutilated by pickling equipment, conveyors or rough handling are to be included in this group.



FIGURE 76. *Improperly formed or mutilated—acceptable.*

Note the occasional overlay of metal at the cup mouth interior on small cups like these. This overlay, if known not to be lamination, is acceptable.

•

Laminations—Unacceptable. (Not illustrated.) (See fig. 6.)

•

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 77. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives.



FIGURE 78. *Scratches—acceptable.*

Light scratches such as those illustrated are acceptable.



FIGURE 79. *Oil, grease, dirt—unacceptable.*



FIGURE 80. *Oil, grease, dirt—acceptable.*



FIGURE 81. *Oxidation—unacceptable.*

Oxidation includes discoloration because of failure in pickling and rinsing to remove the oxides formed in anncaling.

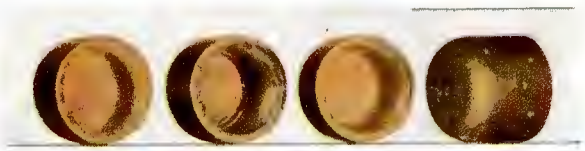


FIGURE 82. *Oxidation—acceptable.*



FIGURE 83. *Dents or bent edges—unacceptable.*



FIGURE 84. *Dents or bent edges—acceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .30 GILDING METAL CLAD STEEL BULLET JACKET CUPS



FIGURE 85. *Defective cladding—unacceptable.*

Defective cladding includes wire edges, scaly metal, and reversed cladding as well as cladding which may be peeling or which has been drawn up into a wire edge. Any cup which appears, without magnification, to be scaly shall be considered a defective. Wire edges may occur on either the inner or outer wall of the cup. The fourth cup from the left is an example of a major wire edge. Cups captioned "Scaly Metal" figures 15 and 30 and identified as A, B, C, and A, B, C, respectively, can be used as other examples of unacceptable wire edges; those cups identified as D, E, and D, E, F, figures 16 and 31, respectively, are acceptable.



FIGURE 86. *Improperly formed or mutilated—unacceptable.*



FIGURE 87. *Improperly formed or mutilated—unacceptable.*



FIGURE 88. *Improperly formed or mutilated—acceptable.*

•

Laminations—Unacceptable. (Not illustrated.) (See fig. 6.)

•



FIGURE 89. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 90. *Scratches—acceptable.*

Scratches no worse than those shown here, whether they be on the interior or the exterior of the cup, are acceptable.



FIGURE 91. *Oil, grease, dirt—unacceptable.*



FIGURE 92. *Oil, grease, dirt—acceptable.*

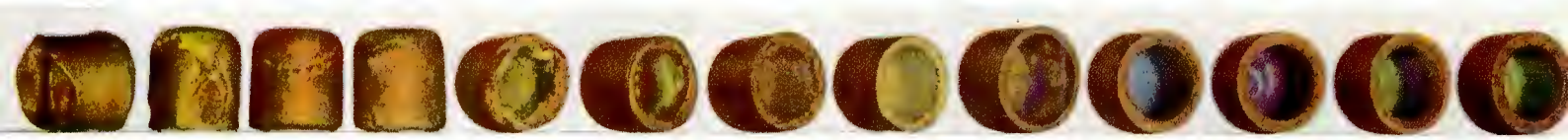


FIGURE 93. *Oxidation—unacceptable.*

Oxidation includes discoloration because of failure in pickling and rinsing to remove the oxides formed in annealing.

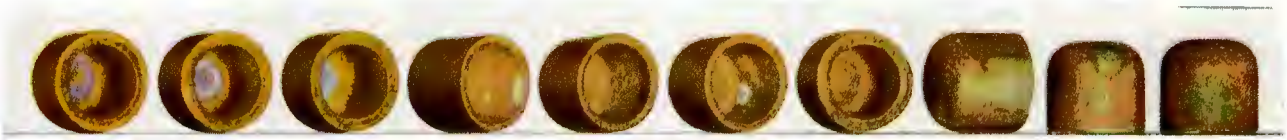


FIGURE 94. *Oxidation—acceptable.*



FIGURE 95. *Dents or bent edges—unacceptable.*



FIGURE 96. *Dents or bent edges—acceptable.*

If cups are not dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .30 CARBINE GILDING METAL BULLET JACKET CUPS



FIGURE 97. *Scaly metal—unacceptable.*

Any cup which appears, without magnification, to be scaly is to be counted as a defective. Scaly metal includes slivers, blisters, cracks, and wire edges. Wire edges on the inner or outer wall of the cup (not shown above) shall be unacceptable.



FIGURE 98. *Improperly formed or mutilated—unacceptable.*

Cups having "V'd" edges, spills, etc., as well as those mutilated by pickling equipment, conveyors or rough handling are to be included in this group.



FIGURE 99. *Improperly formed or mutilated—acceptable.*

Spills no deeper than the "V'd" edges shown are not to be counted as defectives.

•

Laminations—unacceptable. (Not illustrated.)
(See Fig. 6.)

•



FIGURE 100. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives.

FIGURE 101. *Scratches—acceptable.*

Light scratches on the inside or outside base which fade out in the sidewall, and light inside sidewall scratches which do not carry across the base are acceptable if they have clean edges. Draw scratches which cannot be detected by the fingernail are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 102. *Oil, grease, dirt—unacceptable.*



FIGURE 103. *Oil, grease, dirt—acceptable.*

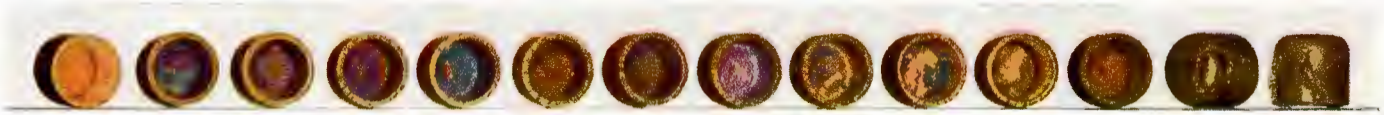


FIGURE 104. *Oxidation—unacceptable.*

A good cup has been included at the left for comparison. Note the two cups at right from which metal has been removed by electrolytic action in pickling.



FIGURE 105. *Oxidation—acceptable.*

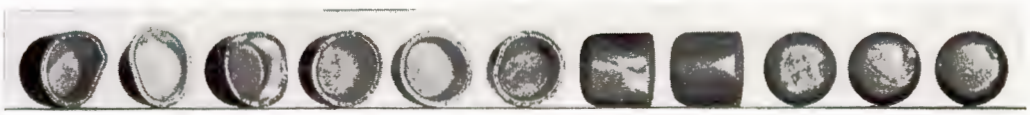


FIGURE 106. *Dents or bent edges—unacceptable.*



FIGURE 107. *Dents or bent edges—acceptable.*

If cups are not dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .45 STEEL CASE CUPS Standards Applicable also to CALIBER .30 CARBINE STEEL CASE CUPS



FIGURE 108. *Scratches—unacceptable*
Cups having deep scratches such as those shown are defectives.



FIGURE 109. *Scratches—acceptable*
Light scratches such as those illustrated are acceptable.



FIGURE 110. *Improperly Formed or Mutilated—unacceptable*
This group includes having "V'd" edges, spills, etc., as well as those mutiliated by rough handling or equipment. Note also the inverse bevel on the cup at the extreme right end.



FIGURE 111. *Improperly Formed or Mutilated*
Small spills like those shown are acceptable. "V'd" edges like that shown shall be unacceptable.

•

Laminations—unacceptable (Not illustrated.) (See Fig. 6.)

•



FIGURE 112. *Scaly Metal—unacceptable.*
A cup which appears, without magnification, to have been scaly before it was copper coated is to be counted as defective. Scaly metal includes wire edges such as those shown on the first three cups at the left, slivers, blisters, and cracks as well as coating which has been drawn up into a wire edge. Wire edges may occur on either the inner or outer wall of the cup. Cups captioned "Scaly Metal" figure 50 and identified as A, B, C can be used as other examples of unacceptable wire edges; those cups identified as D, E, F, figure 51 are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

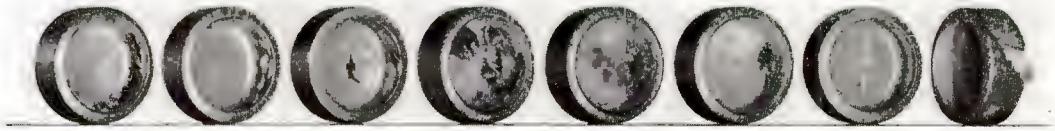


FIGURE 113. *Nonadherent copper coating—unacceptable.*

A cup from which the copper coating is peeling or on which the coating has blistered is to be counted as a defective. Note examples of acceptable degree of defect shown in figure 53.



FIGURE 114. *Oil, grease, dirt—unacceptable.*



FIGURE 115. *Oil, grease, dirt—acceptable.*



FIGURE 116. *Oxidation—unacceptable.*

Discoloration like that illustrated, caused by failure in pickling and rinsing to remove coating formed in annealing or when hot cups are dipped in pickling solution, is unacceptable.



FIGURE 117. *Oxidation—acceptable.*



FIGURE 118. *Dents or bent edges—unacceptable.*



FIGURE 119. *Dents or bent edges—acceptable.*

If cups are not dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .45 GILDING METAL CLAD AND COPPER PLATED STEEL BULLET JACKET CUPS



FIGURE 120. *Defective cladding or plating—unacceptable.*

Defective cladding or plating includes wire edges, scaly metal. cracks, and reversed cladding, as well as cladding or plating which may be peeling or which has been drawn up into a wire edge. Any cup which appears, without magnification, to be scaly shall be considered a defective. Wire edges may occur on either the inner or outer wall of the cup. Cups captioned "Scaly Metal" figures 15 and 30 and identified as A, B, C, and A, B, C, respectively, can be used as examples of unacceptable wire edges; those cups identified as D, E, and D, E, F, figures 16 and 31, respectively, are acceptable.



FIGURE 121. *Improperly formed or mutilated—unacceptable.*



FIGURE 122. *Improperly formed or mutilated—acceptable.*

•

Laminations—unacceptable. (Not illustrated.)
(See fig. 6.)

•



FIGURE 123. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives.



FIGURE 124. *Scratches—acceptable.*

Scratches no worse than those shown here, whether they be on the interior or exterior of the cup, are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 125. *Oil, grease, dirt—unacceptable.*



FIGURE 126. *Oil, grease, dirt—acceptable.*



FIGURE 127. *Oxidation—unacceptable.*

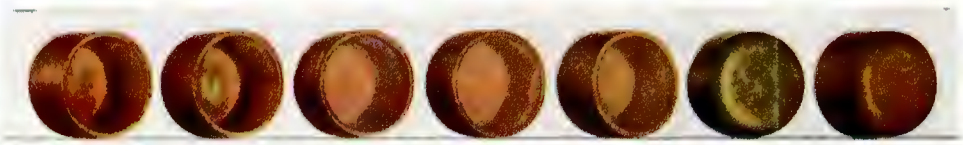


FIGURE 128. *Oxidation—acceptable.*



FIGURE 129. *Dents or bent edges—unacceptable.*



FIGURE 130. *Dents or bent edges—acceptable.*

If cups are not dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .50 BRASS CASE CUPS (THREE DRAW)

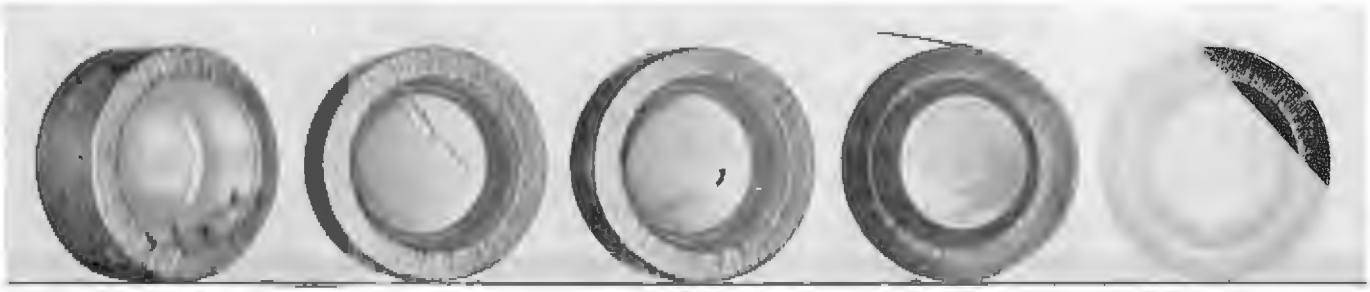


FIGURE 131. *Scaly metal—unacceptable.*



[----------Unacceptable----------] [----------Acceptable----------]

FIGURE 132. *Scaly metal.*

Any cup which appears, without magnification, to be scaly is to be counted as a defective. Scaly metal includes slivers, blisters, cracks, and wire edges. Wire edges may occur on either the inner or outer wall of the cup. The two cups identified as A, B above, and in addition, cups captioned "Scaly Metal" figure 146 and 229 identified as A, and A, B, respectively, are examples of unacceptable wire edges. Cups captioned "Scaly Metal" figure 147 and 230 identified as B, C, and C, D, E, respectively, are acceptable.

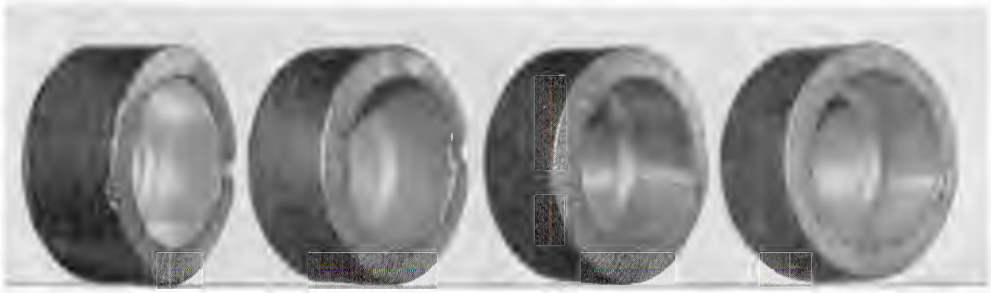


FIGURE 133. *Improperly formed or mutilated—unacceptable.*

Cups having "V'd" edges, spills, etc., as well as those mutilated by pickling equipment, conveyors or rough handling are to be included in this group.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

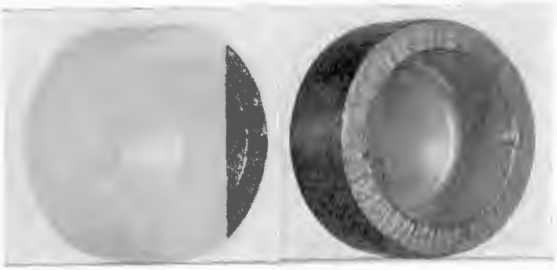


FIGURE 134. *Improperly formed or mutilated—acceptable*

•

Laminations—Unacceptable (Not illustrated.) (See fig. 6.)

•

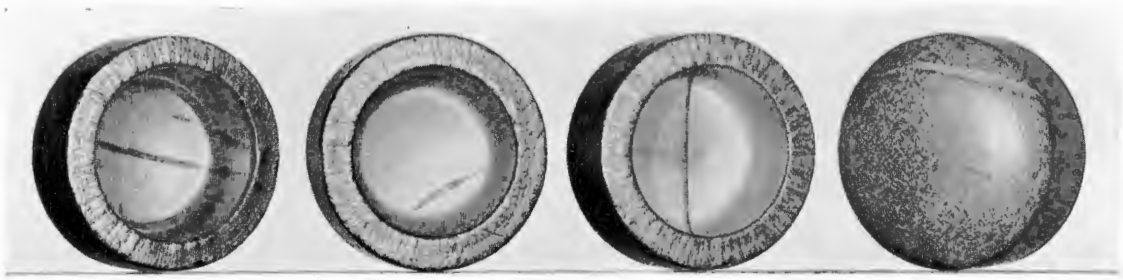


FIGURE 135. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives. Note gouge-like scratch in cup at right end of group.



FIGURE 136. *Scratches—acceptable.*

Light scratches on the interior or exterior base which fade out in the sidewall and light interior sidewall scratches which do not carry across the base are acceptable if they have clean edges. Light exterior draw scratches are acceptable (see acceptable scratches Figure 65).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 137. *Oil, grease, dirt—unacceptable.*



FIGURE 138. *Oil, grease, dirt—acceptable.*

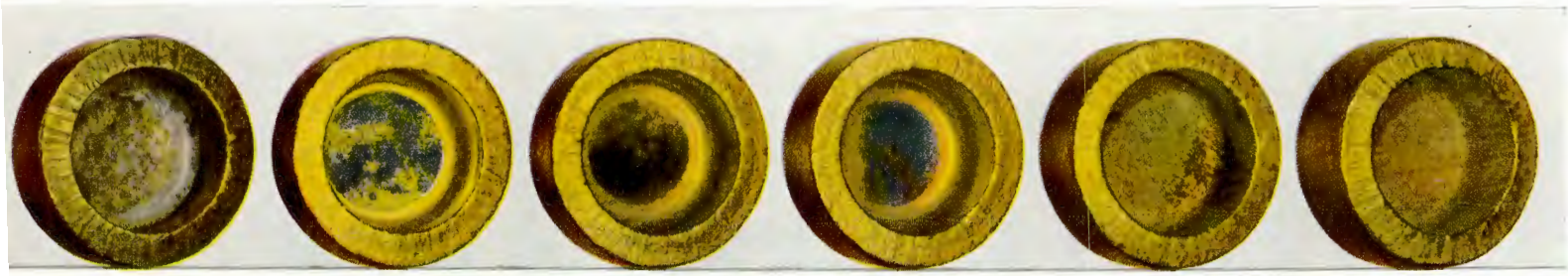


FIGURE 139. *Oxidation—unacceptable.*

Discoloration like that illustrated, caused by failure in pickling and rinsing to remove coatings formed in annealing or when hot cups are immersed in pickling solution, is unacceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

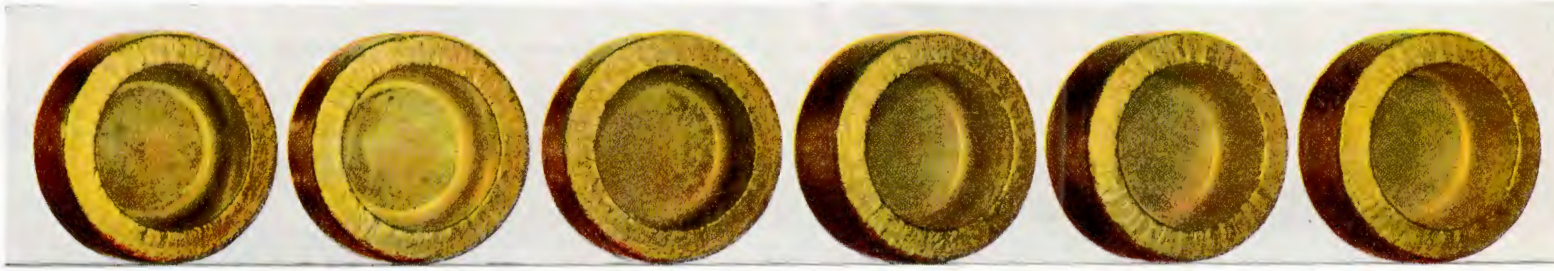


FIGURE 140. *Oxidation—acceptable.*

The bright yellow color produced by over-pickling is to be considered acceptable.



FIGURE 141. *Red spots—unacceptable.*

FIGURE 142. *Red spots—acceptable.*

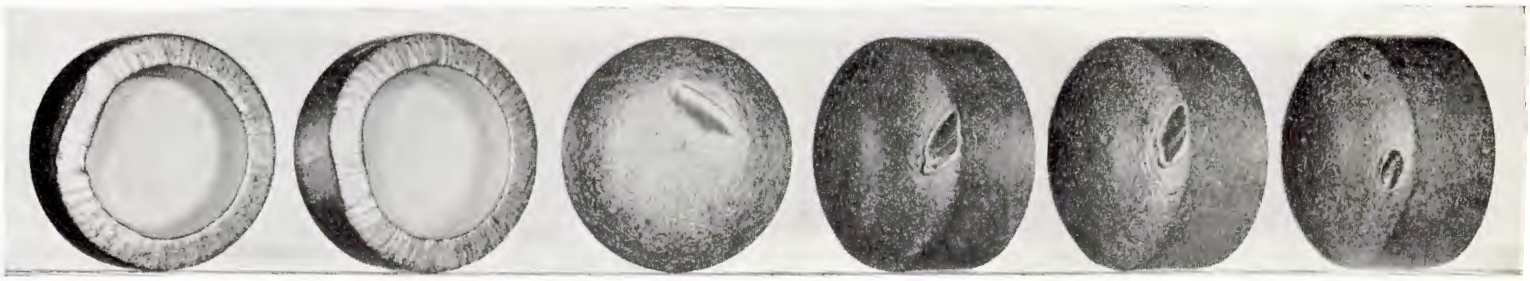


FIGURE 143. *Dents or bent edges—unacceptable.*



FIGURE 144. *Dents or bent edges—acceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .50 BRASS CASE CUPS (TWO DRAW)



FIGURE 145. *Scaly metal—unacceptable.*

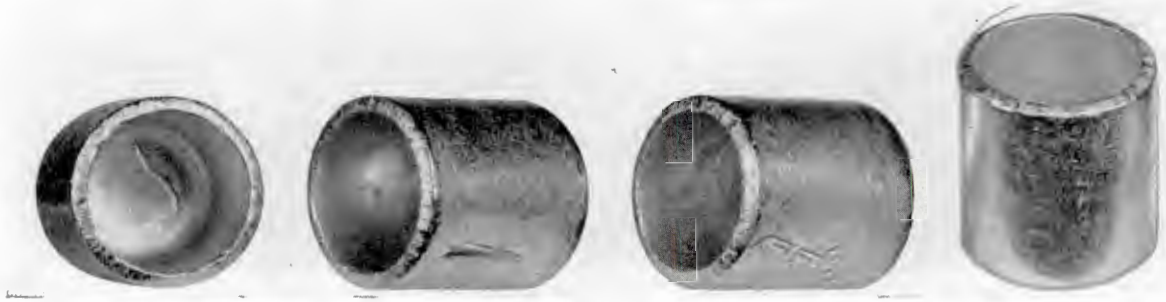


FIGURE 146. *Scaly metal—unacceptable.*



FIGURE 147. *Scaly metal—acceptable.*

Any cup which appears, without magnification, to be scaly is to be considered defective. Scaly metal includes slivers, blisters, wire edges, and cracks. Wire edges may occur on either the inner or outer wall of the cup. Cup identified as A is an example of an unacceptable wire edge; cups identified B and C are acceptable. Note rough orangepeel effect on cup second from right end, bottom picture.



FIGURE 148. *Improperly formed or mutilated—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 149. *Improperly formed or mutilated—acceptable.*

•

Laminations—unacceptable. (Not illustrated.) (See fig. 6.)

•



FIGURE 150. *Scrap—unacceptable.*

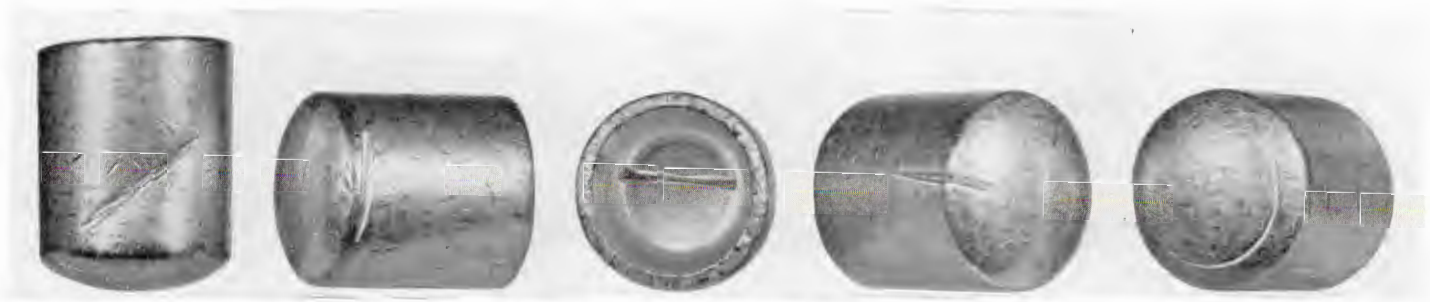


FIGURE 151. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives.

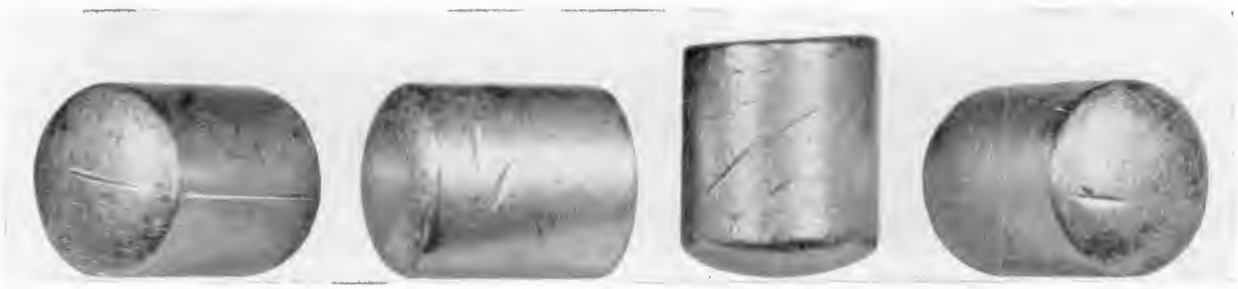


FIGURE 152. *Scratches—acceptable.*

Light exterior draw scratches are acceptable (see acceptable scratches fig. 237).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 153. *Oil, grease, dirt—unacceptable.*

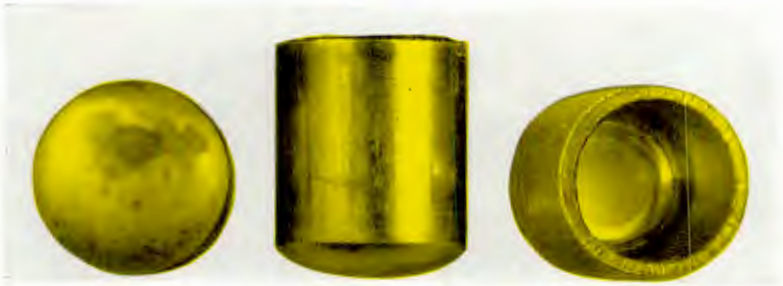


FIGURE 154. *Oil, grease, dirt—acceptable.*

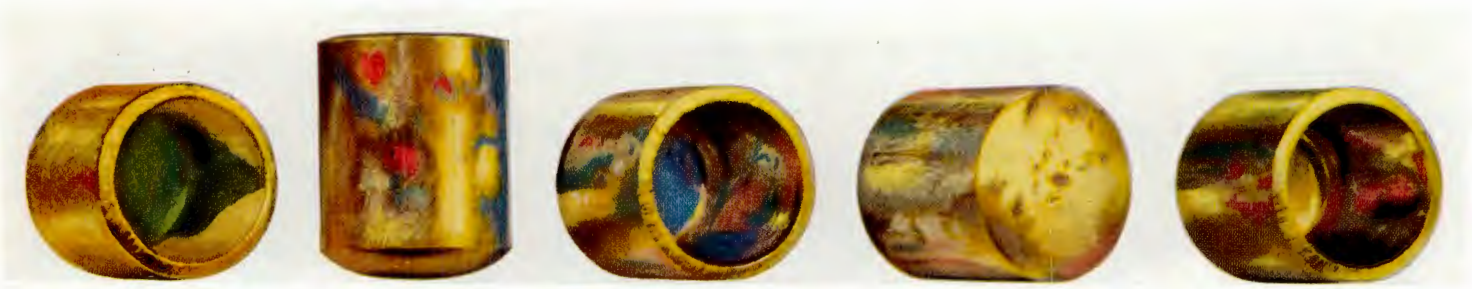


FIGURE 155. *Oxidation—unacceptable.*

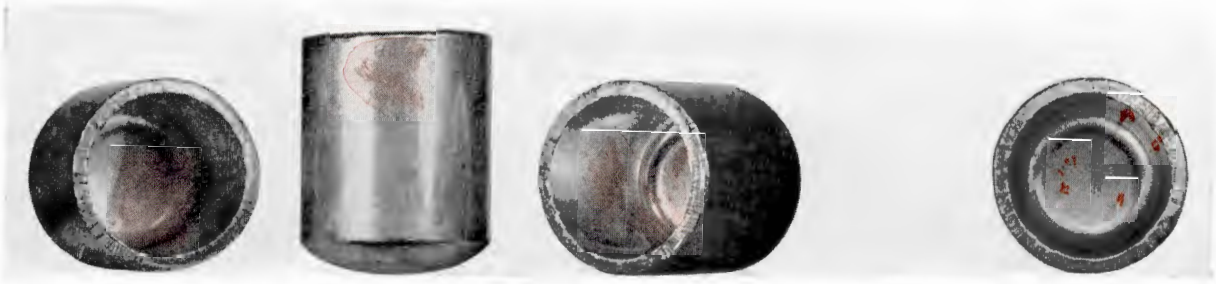
Discoloration like that illustrated, caused by failure in pickling and rinsing to remove coatings formed in annealing or when hot cups are immersed in pickling solution, is unacceptable.



FIGURE 156. *Oxidation—acceptable.*

The bright yellow color produced by overpickling is to be considered acceptable oxidation, even though it is undesirable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



[— — — — — — — — — Unacceptable — — — — — — — — —]

[Acceptable]

FIGURE 157. *Red spots.*

Defectives are cups having copper deposited on them due to electrochemical action in pickling. Such action generally occurs when free iron is present. Black specks in the red spots indicate iron inclusions. Cups with such inclusions and those with large areas of red discoloration are defectives.

Cups are not defectives where pickling action has removed inclusions, or where area affected is no greater than that shown.

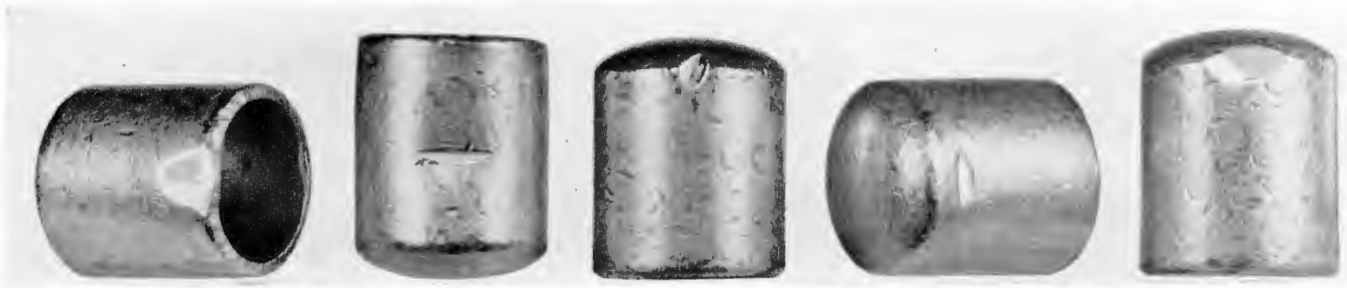


FIGURE 158. *Dents or bent edges—unacceptable.*



FIGURE 159. *Dents or bent edges—acceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .50 STEEL CASE CUP DISKS



FIGURE 160. *Seams—unacceptable.*



FIGURE 161. *Seams—acceptable.*

Light seams which cannot be detected by the fingernail are acceptable.



FIGURE 162. *Improperly formed or mutilated—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 163. *Improperly formed or mutilated—acceptable.*

Laminations—unacceptable. (Not illustrated.) (See fig. 6.)



FIGURE 164. *Scrap—unacceptable.*



FIGURE 165. *Nicks, cuts and scratches—unacceptable.*



FIGURE 166. *Nicks, cuts and scratches—acceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

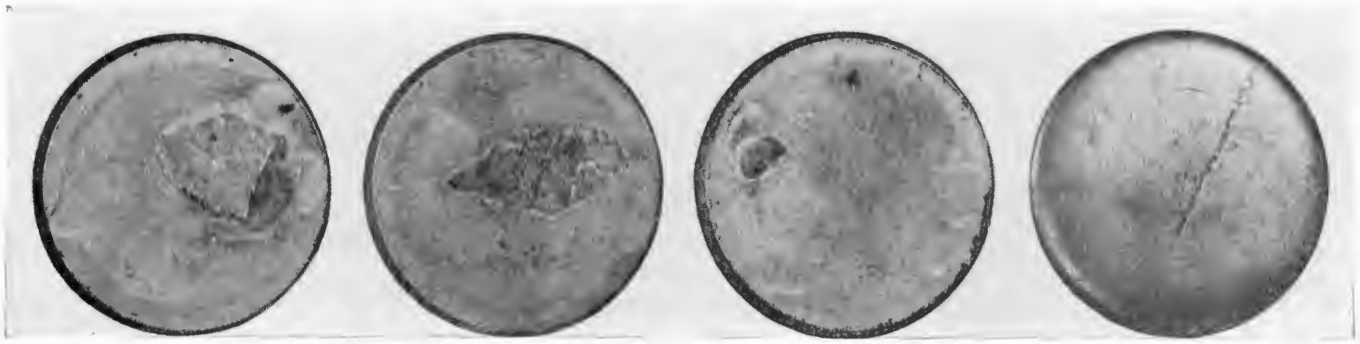


FIGURE 167. *Scale—unacceptable.*



FIGURE 168. *Scale—unacceptable.*

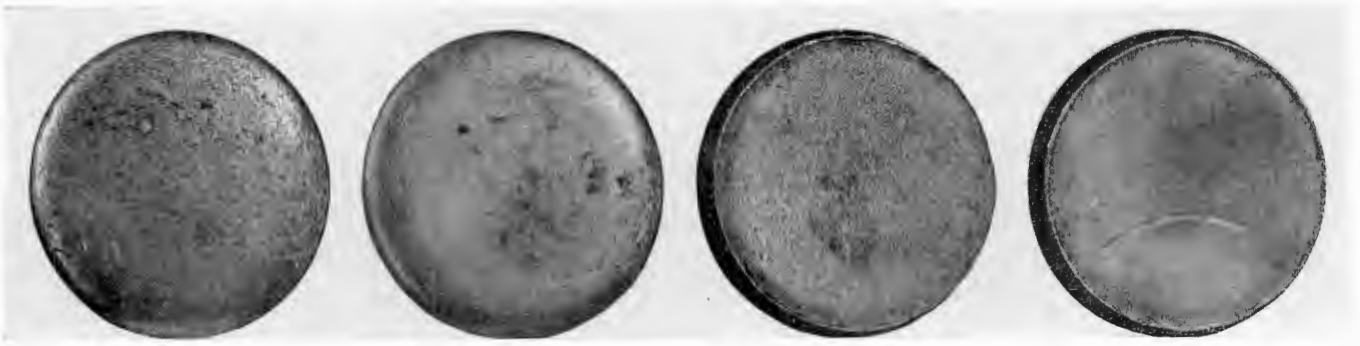


FIGURE 169. *Scale—acceptable.*

Any disk which appears, without magnification, to be scaly is to be counted a defective. Scaly metal includes slivers, blisters, wire edges, and cracks. The disk identified as A is an example of an unacceptable wire edge condition; a wire edge no greater than that shown on the disk identified as B is acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 170. *Oil, grease, dirt—unacceptable.*



FIGURE 171. *Oil, grease, dirt—acceptable.*



FIGURE 172. *Oxidation—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 173. *Oxidation—acceptable.*

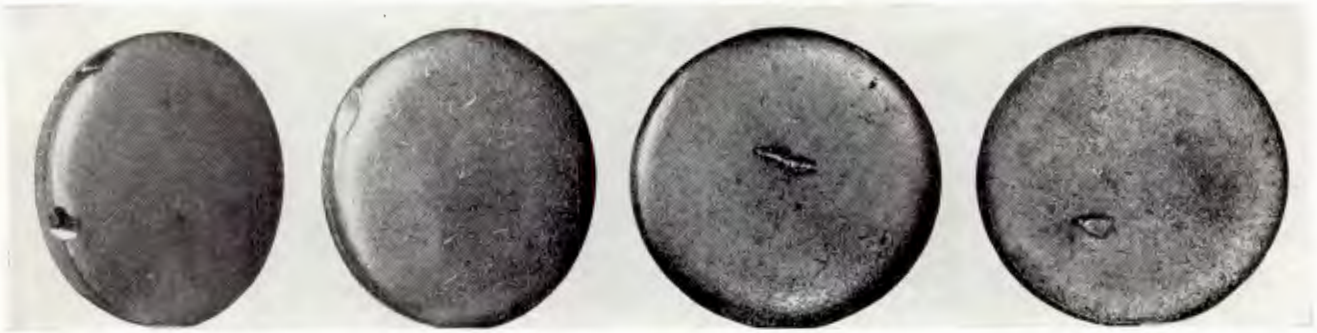


FIGURE 174. *Dents or bent edges—unacceptable.*



FIGURE 175. *Dents or bent edges—acceptable.*

If disks are not dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .50 STEEL CASE CUPS

(The cups illustrated are presumed to have been phosphate coated, after cupping from uncoated disks.)



FIGURE 176. *Scratches—unacceptable.*

The defect includes handling and draw scratches, and seams. Cuts having deep scratches such as those shown are defectives.



FIGURE 177. *Scratches—acceptable.*

Light handling and draw scratches and light seams which cannot be detected by the fingernail are acceptable.



FIGURE 178. *Improperly formed or mutilated—unacceptable.*

This group includes cups having "V'd" edges, spills, etc., as well as those mutilated by pickling equipment, conveyors or rough handling.



FIGURE 179. *Improperly formed or mutilated.*

Improperly formed or mutilated cups like those shown in left and center photographs above are acceptable. "V'd" edges like that shown in the right photograph above shall be unacceptable.

•

Laminations—unacceptable. (Not illustrated.) (See Fig. 6.)

•

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 180. *Scrap—unacceptable*

Note double cup second from right end.

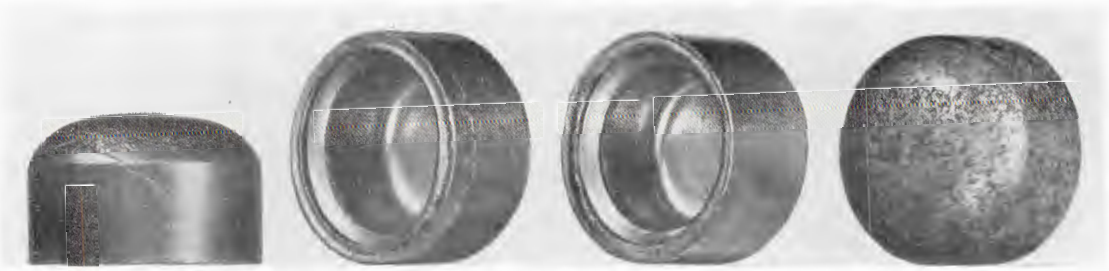


FIGURE 181. *Scaly metal—unacceptable.*



FIGURE 182. *Scaly metal—unacceptable.*

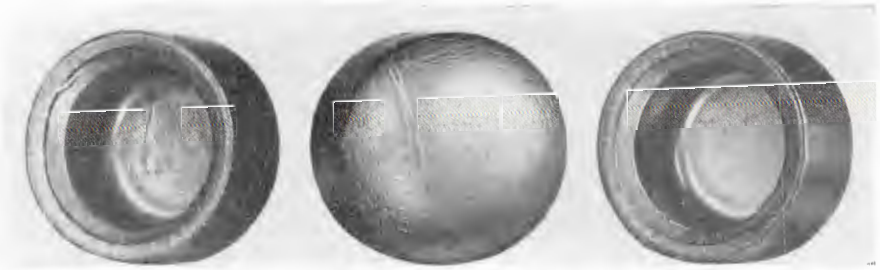


FIGURE 183. *Scaly metal—acceptable.*

Any cup which appears, without magnification, to have been scaly before it was phosphate coated is to be counted a defective. Scaly metal includes slivers, wire edges, etc. Wire edges may occur on either the inner or outer wall of the cup. The cup identified as A is an example of an unacceptable wire edge; those cups identified as B and C are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

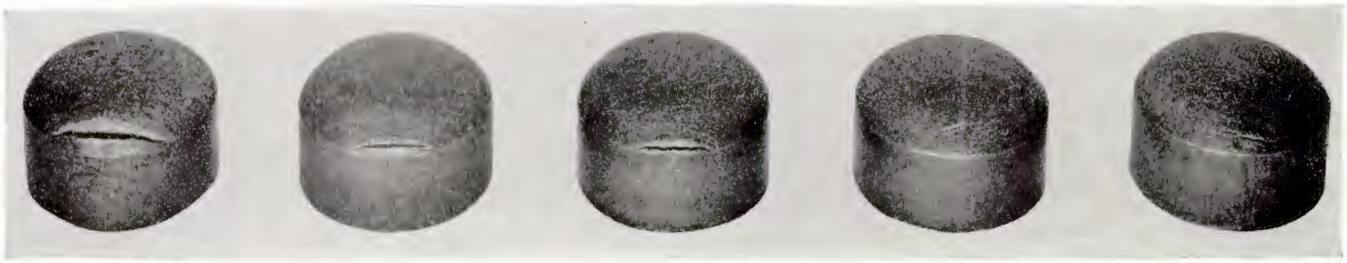


FIGURE 184. *Metal separation—unacceptable.*



FIGURE 185. *Oil, grease, dirt—unacceptable.*



FIGURE 186. *Oil, grease, dirt—acceptable.*

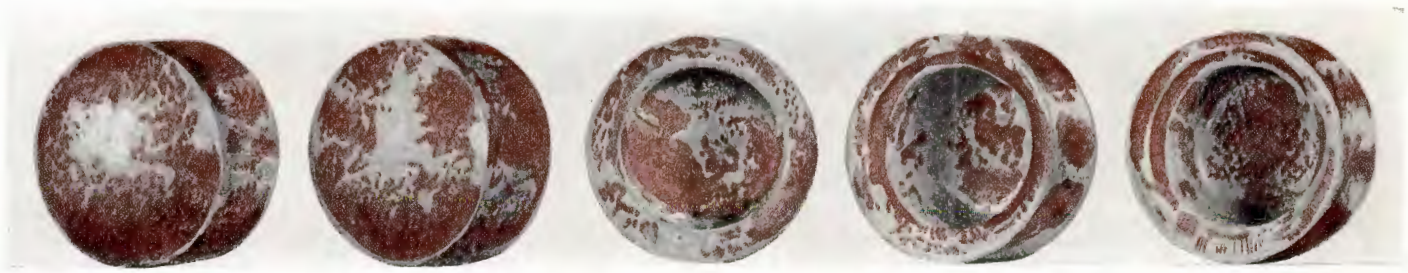


FIGURE 187. *Oxidation—unacceptable.*

Oxidation includes discoloration because of failure in pickling and rinsing to remove the oxides formed in annealing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 188. *Oxidation—acceptable.*

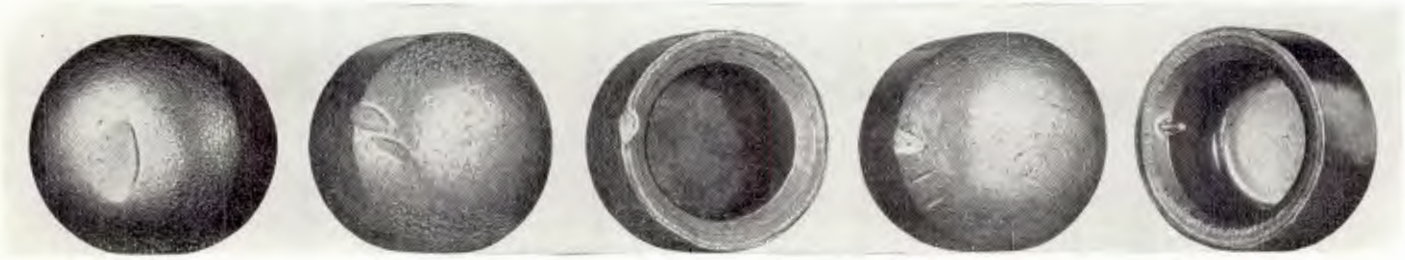


FIGURE 189. *Dents or bent edges—unacceptable.*



FIGURE 190. *Dents or bent edges—acceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .50 GILDING METAL BULLET JACKET CUPS



FIGURE 191. *Scaly metal—unacceptable.*



FIGURE 192. *Scaly metal—unacceptable.*

Any cup which appears, without magnification, to be scaly is to be considered a defective. Scaly metal includes blisters, wire edges, cracks and slivers (see cup third from right in upper photo). Wire edges on the inner or outer wall of the cup (not shown above) shall be unacceptable.

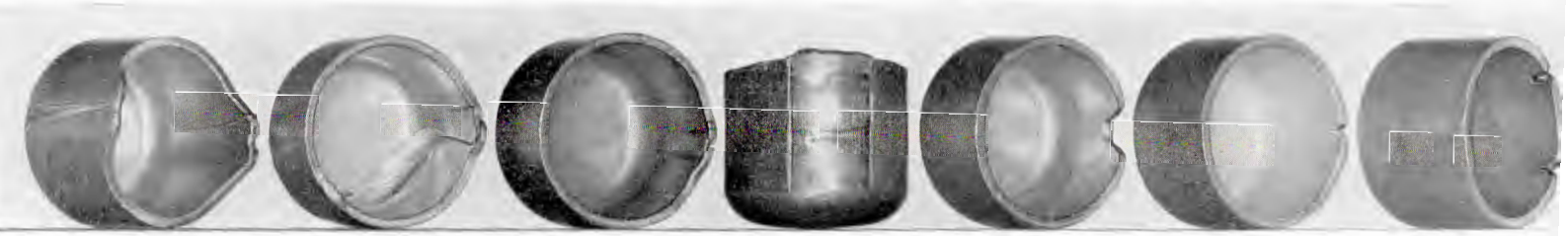


FIGURE 193. *Improperly formed or mutilated—unacceptable.*

Cups having "V'd" edges, spills, etc., as well as those mutilated by pickling equipment, conveyors or rough handling are to be included in this group.



FIGURE 194. *Improperly formed or mutilated—acceptable.*

Laminations—Unacceptable. (Not illustrated.) (See Fig. 6.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 195. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives.



FIGURE 196. *Scratches—acceptable.*

Scratches no worse than those shown, on the interior or exterior of the cup, are acceptable.



FIGURE 197. *Oil, grease, dirt—unacceptable.*



FIGURE 198. *Oil, grease, dirt—acceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

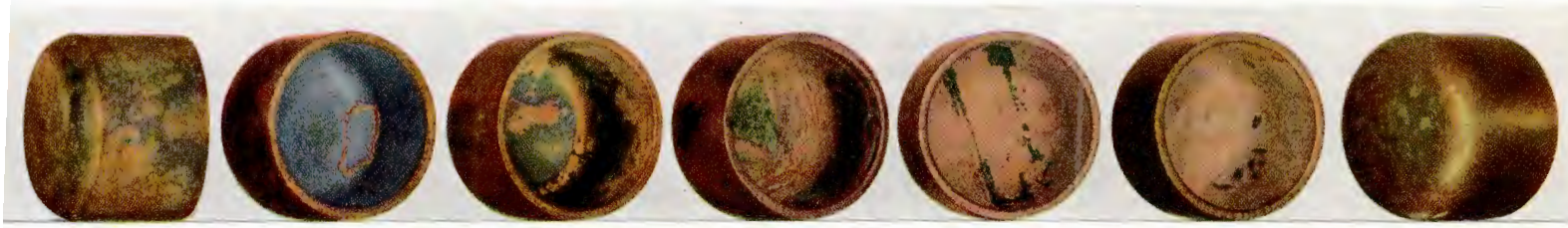


FIGURE 199. *Oxidation—unacceptable.*

Oxidation includes discoloration caused by failure in pickling and rinsing to remove the oxides formed in annealing.

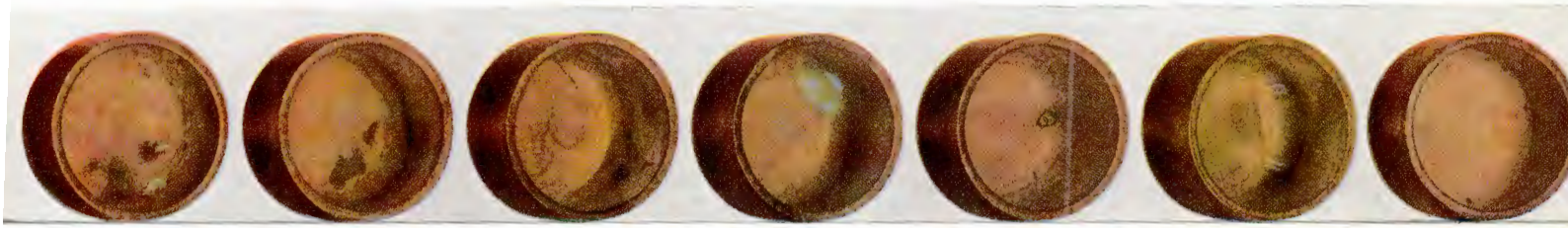


FIGURE 200. *Oxidation—acceptable.*

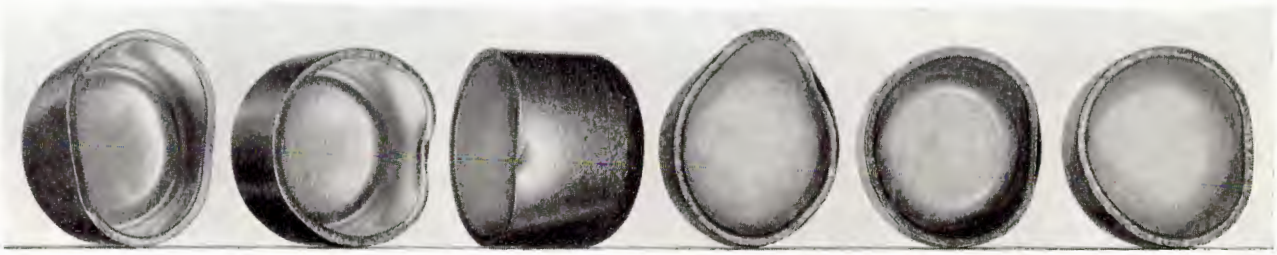


FIGURE 201. *Dents or bent edges—unacceptable.*

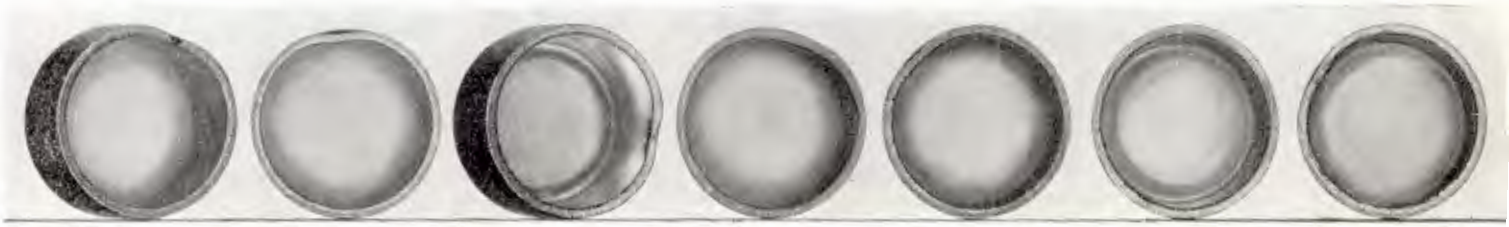


FIGURE 202. *Dents or bent edges—acceptable.*

If cups are not bent out-of-round or dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

CALIBER .50 GILDING METAL CLAD STEEL BULLET JACKET CUPS



FIGURE 203. *Defective cladding—unacceptable.*

Defective cladding includes wire edges, scaly metal and reversed cladding as well as cladding which may be peeling or which has been drawn up into a wire edge. Any cup which appears, without magnification, to be scaly shall be considered a defective. Wire edges may occur on either the inner or outer wall of the cup. Cups captioned "Scaly Metal" Figures 15 and 30 and identified A, B, C, and A, B, C, respectively, can be used as examples of unacceptable wire edges; those cups identified as D, E, and D, E, F, Figures 16 and 31, respectively, are acceptable.



FIGURE 204. *Improperly formed or mutilated—unacceptable.*

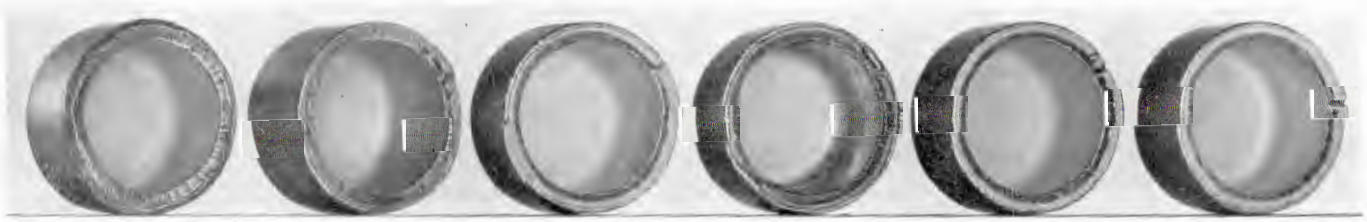


FIGURE 205. *Improperly formed or mutilated—acceptable.*

Laminations—unacceptable. (Not illustrated.) (See fig. 6.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 206. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives.

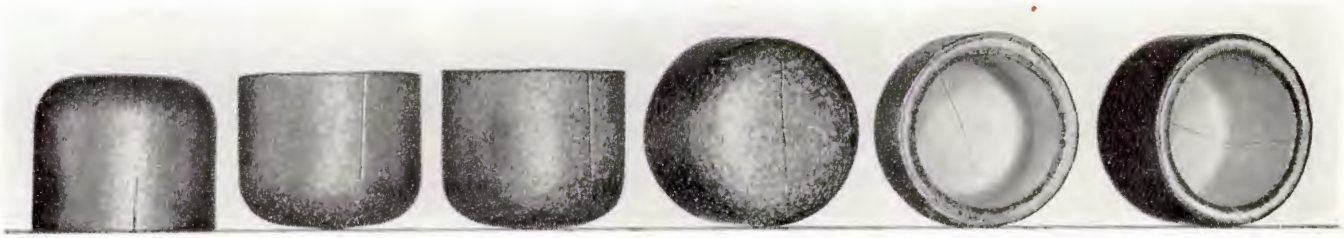


FIGURE 207. *Scratches—acceptable.*

Scratches no worse than those shown on the interior or the exterior of the cup, as well as light draw scratches which cannot be detected by the fingernail, are acceptable.



FIGURE 208. *Oil, grease, dirt—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 209. *Oil, grease, dirt—acceptable.*



FIGURE 210. *Oxidation—unacceptable.*

Oxidation includes discoloration caused by failure in pickling and rinsing to remove the oxides formed in annealing.

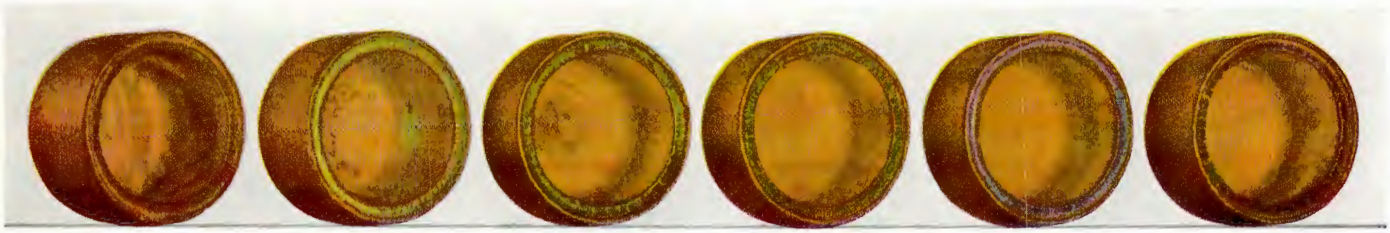


FIGURE 211. *Oxidation—acceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 212. *Dents, or bent edges—unacceptable.*

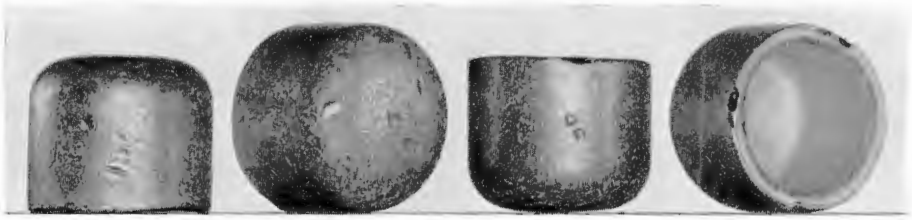


FIGURE 213. *Dents or bent edges—acceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 214. *Scaly metal—unacceptable.*



FIGURE 215. *Scaly metal—acceptable.*

Any disk which appears, without magnification, to be scaly is to be counted as a defective. Scaly metal includes slivers, blisters, wire edges, and cracks. The disks identified as A, B are examples of the unacceptable wire edge condition; wire edges no greater than shown on those disks identified as C, D are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 216. *Improperly formed or mutilated—unacceptable.*



FIGURE 217. *Improperly formed or mutilated—acceptable.*

•

Laminations—unacceptable. (Not illustrated.) (See Fig. 6.)

•

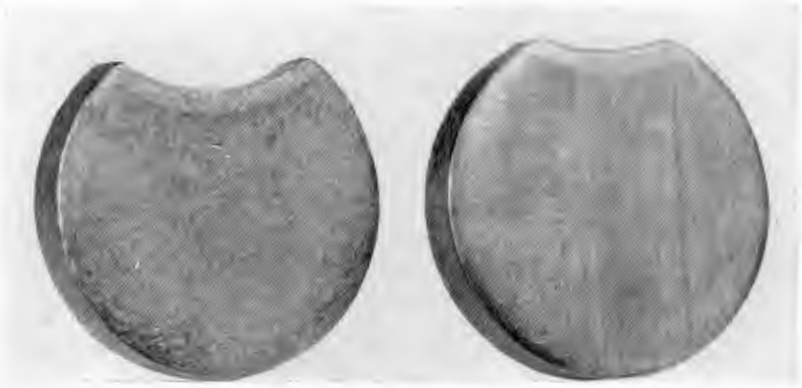


FIGURE 218. *Scrap—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

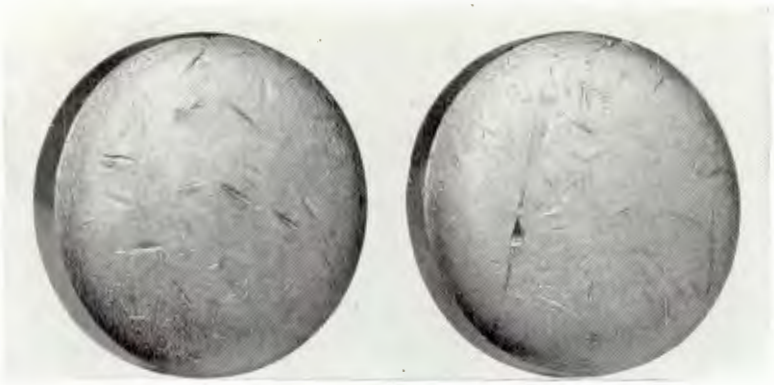


FIGURE 219. *Scratches, cuts or nicks—unacceptable.*



FIGURE 220. *Scratches, cuts or nicks—acceptable.*

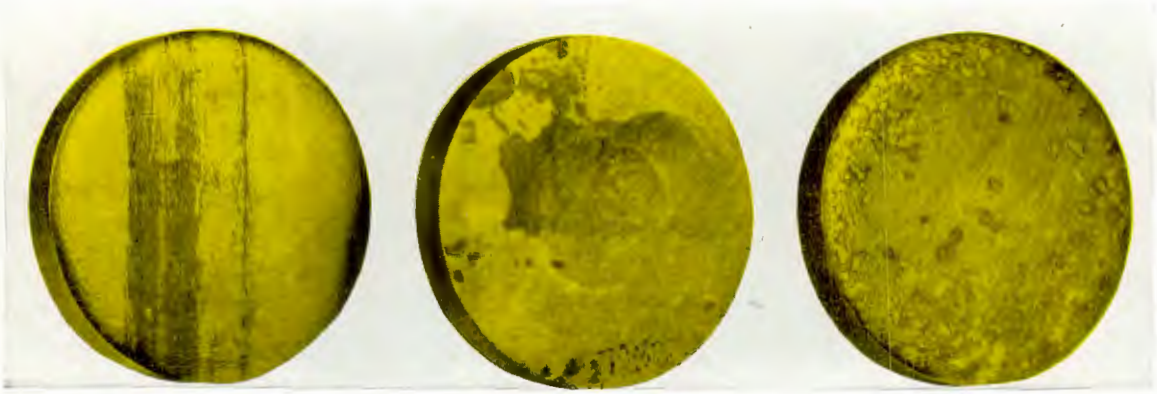


FIGURE 221. *Oil, grease, dirt—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

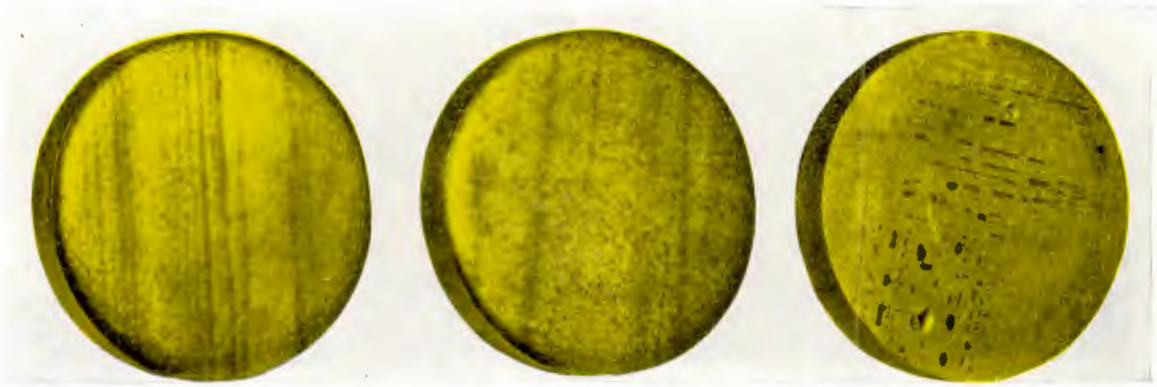


FIGURE 222. *Oil, grease, dirt—acceptable.*



FIGURE 223. *Oxidation—unacceptable.*

Discoloration like that illustrated, caused by failure in pickling and rinsing to remove coatings formed in annealing or when hot disks are immersed in pickling solution, is unacceptable.



FIGURE 224. *Oxidation—acceptable.*

The bright yellow color produced by over-pickling is to be considered acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



[— — — — — — — — — Unacceptable — — — — — — — — —]

[— — — Acceptable — — —]

FIGURE 225. *Red Spots.*

Defectives are disks having copper deposited on them due to electro-chemical action in pickling. Such action generally occurs when free iron is present. Black specks in the red spots indicate iron inclusions. Disks with such inclusions and those with large areas of red discoloration are defectives.

Disks are not defectives where pickling action has removed inclusions, or where area affected is relatively small.



FIGURE 226. *Dents or bent edges—unacceptable.*



FIGURE 227. *Dents or bent edges—acceptable.*

If disks are not dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

20-MM BRASS CASE CUPS

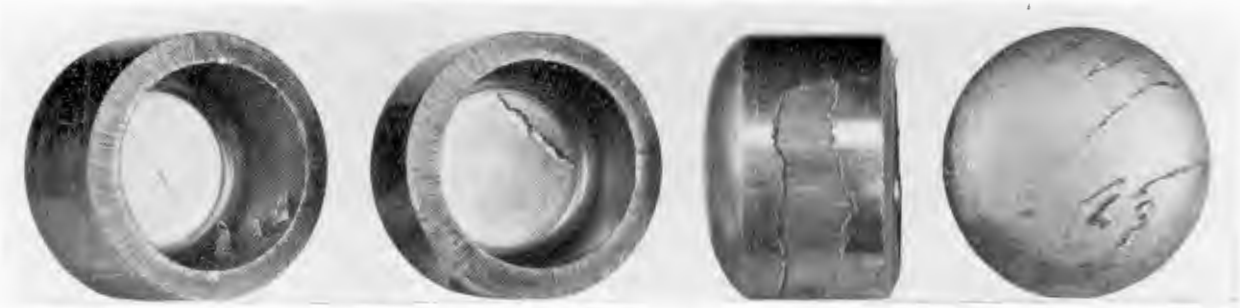


FIGURE 228. *Scaly metal—unacceptable.*



FIGURE 229. *Scaly metal—unacceptable.*

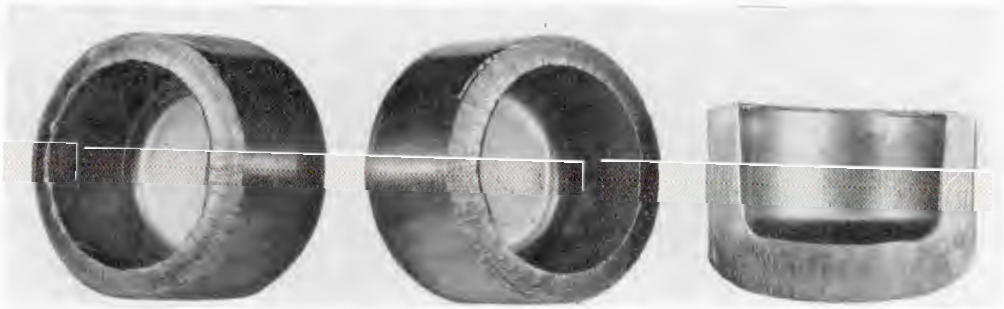


FIGURE 230. *Scaly metal—acceptable.*

Any cup which appears, without magnification, to be scaly is to be counted as a defective. Scaly metal includes slivers, blisters, cracks, and wire edges. Wire edges may occur on the inner or outer wall of the cup. Cups identified as A, B, are examples of unacceptable wire edges; cups identified C, D, E are acceptable.



FIGURE 231. *Improperly formed or mutilated—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 232. *Improperly formed or mutilated—unacceptable.*

Cups having "V'd" edges, spills, etc., as well as those mutilated by pickling equipment, conveyors or rough handling are included in this group.



FIGURE 233. *Improperly formed or mutilated—acceptable.*

•

Laminations—unacceptable. (Not illustrated.) (See fig. 6.)

•



FIGURE 234. *Scrap—unacceptable.*

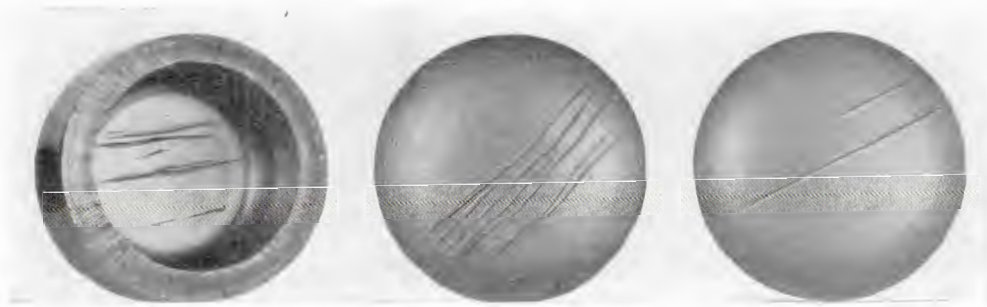


FIGURE 235. *Scratches—unacceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

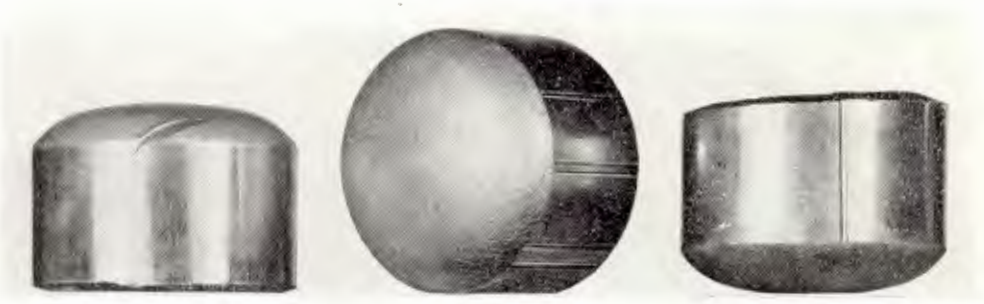


FIGURE 236. *Scratches—unacceptable.*

Cups having deep scratches such as those shown are defectives.

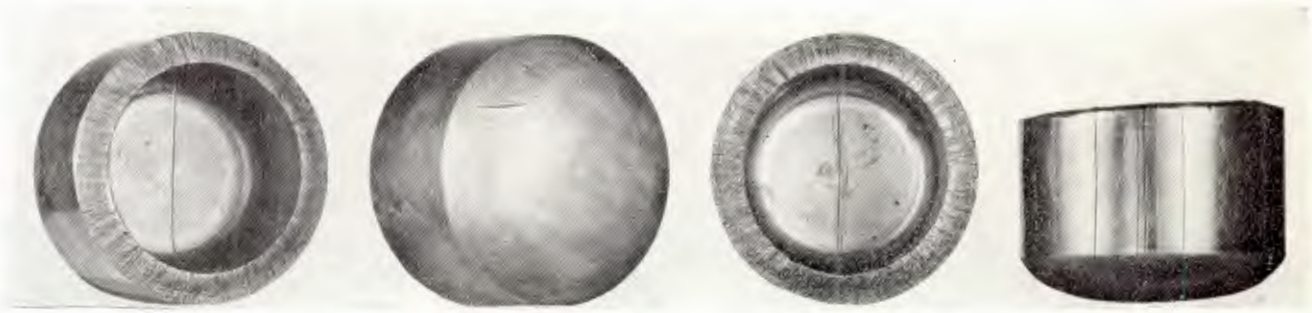


FIGURE 237. *Scratches—acceptable.*

Light scratches on the interior or exterior base which fade out in the sidewall and light interior sidewall scratches which do not carry across the base are acceptable if they have clean edges. Light exterior draw scratches, such as shown on the right, are acceptable.

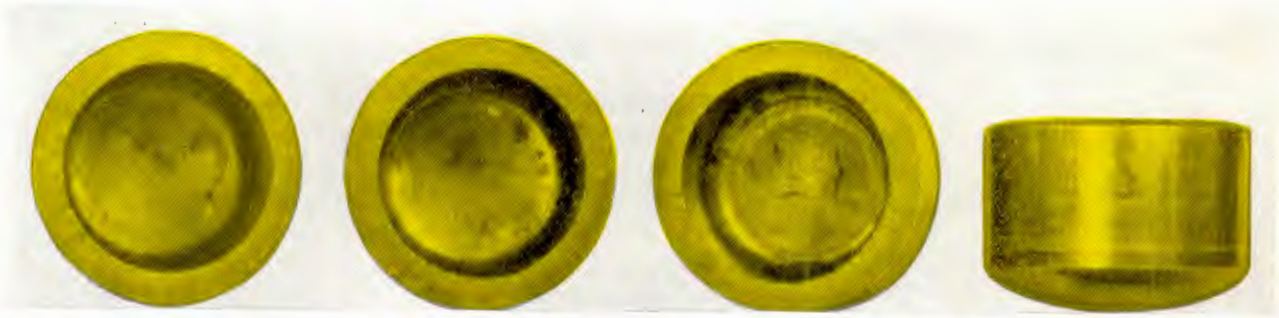


FIGURE 238. *Oil, grease, dirt—unacceptable.*

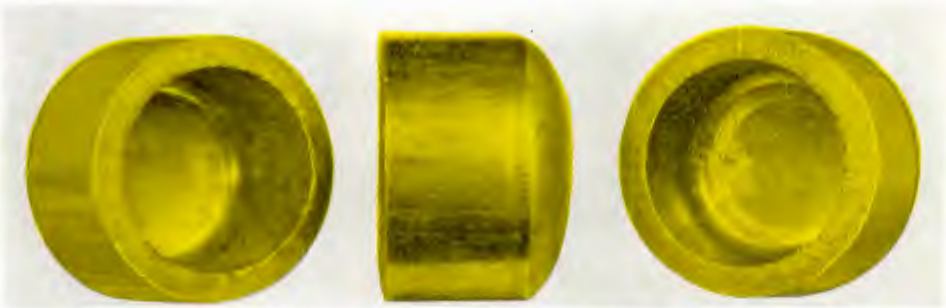


FIGURE 239. *Oil, grease, dirt—acceptable.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.



FIGURE 240. *Oxidation—unacceptable.*

Discoloration like that illustrated, caused by failure in pickling and rinsing to remove coatings formed in annealing or when hot cups are immersed in pickling solution, is unacceptable.



FIGURE 241. *Oxidation—acceptable.*

The bright yellow color produced by overpickling is considered acceptable even though it is undesirable.



FIGURE 242. *Red spots—unacceptable.*

Defectives are cups having copper deposited on them due to electrochemical action in pickling. Such action generally occurs when free iron is present. Black specks in the red spots indicate iron inclusions. Cups with such inclusions and those with large areas of red discoloration are defectives.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

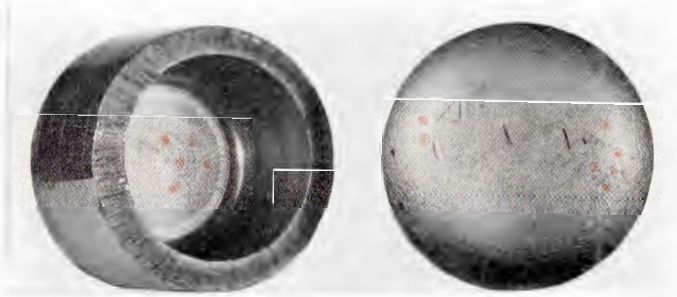


FIGURE 243. *Red spots—acceptable.*

Cups are not defectives where pickling action has removed inclusions, or where area affected is relatively small.

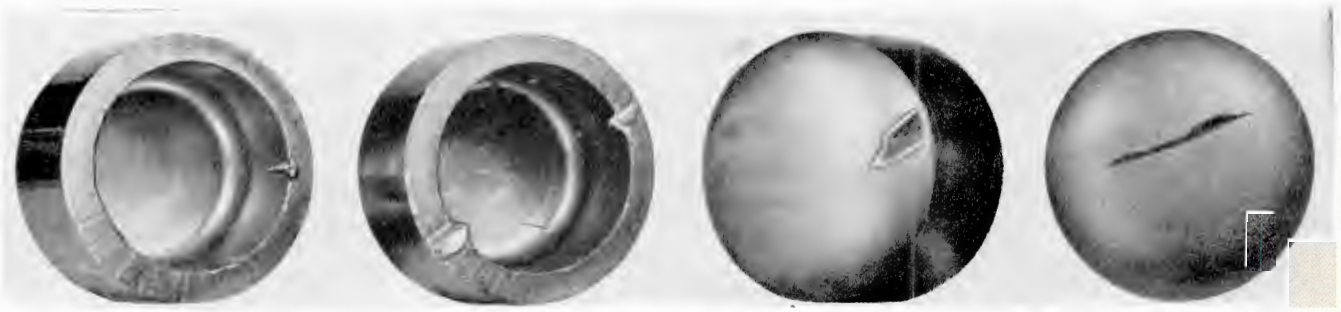


FIGURE 244. *Dents or bent edges—unacceptable.*

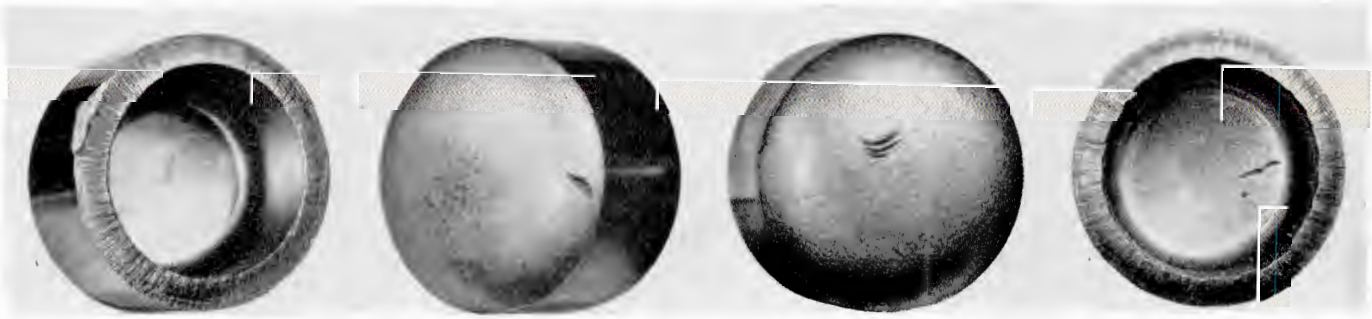


FIGURE 245. *Dents or bent edges—acceptable.*

If cups are not dented more than those shown, they are acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.

Proposed MIL–STD–1232 (ORD)
20 April 1961

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

(Copies of this standard for military use may be obtained as indicated in the foreword to the Index of Military Specifications and Standards.)

(Copies of this standard may be obtained for other than official use by individuals, firms, and contractors from the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D.C.)

(Both the title and identifying symbol number should be stipulated when requesting copies of military standards.

Custodian:
Army—Ord

Preparing activity:
Army—Ord

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:42Z
Check the source to verify that this is the current version before use.